THE COMPLETE BUSINESS KNOWLEDGE ENCYCLOPEDIA

الموسوعة الشاملة للمعرفة في عالم الأعمال

VOLUME 2

How Businesses Work — Accounting and Financial Statements

كيف تعمل الشركات — المحاسبة والقوائم المالية

Every Essential Business Concept, Term, Process, Document, Financial Instrument, and Commercial Practice You Need to Understand the Real Business World

IN THIS VOLUME

- Part II — How Businesses Actually Work
- Part III — Accounting and Financial Statements

A practical reference for finance and computer-science graduates entering the business world · First edition — October 2026

About this volume

This volume is part of a ten-volume reference collection. Chapter numbers run continuously across the collection, so a reference such as “§47.3” identifies the same section in every volume, in the master index (Volume 10), and in the learning roadmap.

Volume	Title
Vol. 1	Foundations and the Language of Business
Vol. 2	**How Businesses Work — Accounting and Financial Statements**
Vol. 3	Corporate Finance, Economics and Business Mathematics
Vol. 4	Banking, International Trade and Treasury
Vol. 5	Commercial Operations — Sales, Supply Chain, Insurance and Risk
Vol. 6	Investment, Capital Markets, Commodities and Market Intelligence
Vol. 7	Law, Entrepreneurship, Management and Technology
Vol. 8	Professional Practice — Communication, Deals and Advanced Thinking
Vol. 9	Document Library, Master Reference, Exercises and Learning Roadmap
Vol. 10	Master Table of Contents, Glossary and Unified Index

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PART II

How Businesses Actually Work

كيف تعمل الشركات فعلياً

Part II looks at the firm as a working system. Part I gave you the words; this part shows the machinery those words describe: how an idea becomes a company, how companies decide what to sell and at what price, how the departments depend on each other, and how a decision in one department ripples through inventory, cash, financing and profit.

The part ends with an industry tour — manufacturing, retail, logistics, software, banking, construction, energy and international trading — because each industry has a characteristic way of making and losing money. Recognising these patterns is one of the fastest ways to sound (and think) like an experienced professional.

LEARNING OBJECTIVES · **Learning objectives for Part II**

- Describe the full lifecycle of a business from opportunity to exit or closure.
- Explain business models, revenue models, pricing strategies and cost structures, and how they determine profitability.
- Trace the effects of one commercial decision across all departments and the financial statements.
- Explain why profitable businesses fail and why high revenue does not guarantee financial health.
- Recognise the economic logic of eight major industries.

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CHAPTER 13

The Business Lifecycle: From Idea to Exit

LEARNING OBJECTIVES · In this chapter

The stages every business passes through — and the questions, documents, risks and financing that dominate each stage.

13.1 The stages at a glance

Stage	Central question	Main activities	Typical financing	Typical failure mode
1. Opportunity	Is there a real problem worth solving?	Observation, research, customer interviews	Founder's own time and savings	Solving a non-problem
2. Validation	Will customers pay, and enough?	MVP, pilots, pre-sales	Savings, friends and family, grants, angels	Building before validating
3. Launch	Can we deliver reliably?	Legal set-up, first hires, operations, first sales	Angels, seed funds, small bank loans with guarantees	Operational chaos, underpricing
4. Growth	Can we scale without breaking?	Systems, management layers, new markets	VC, bank facilities, trade finance, private equity	Running out of working capital
5. Maturity	How do we defend and extract value?	Efficiency, cost control, adjacent products	Bank debt, bonds, retained earnings	Complacency, disruption
6. Renewal or decline	Reinvent, sell or wind down?	Restructuring, M&A, divestment	Restructuring finance, new investors	Too late to act
7. Exit or closure	How is value realised?	Sale, IPO, management buyout, liquidation	Buyer's funding / insolvency process	Value destroyed by delay

بالعربية — دورة حياة المنشأة

تمر المنشأة عادةً بسبع مراحل: اكتشاف الفرصة، ثم التحقق من وجود طلب حقيقي، ثم الإطلاق، ثم النمو، ثم النضج، ثم التجدد أو التراجع، وأخيراً التخارج أو الإغلاق. لكل مرحلة سؤال محوري ومصادر تمويل مناسبة ومخاطر نموذجية. فالشركة الناشئة تموَّل غالباً من المؤسسين والمستثمرين الملائكة، بينما تعتمد الشركة الناضجة على القروض المصرفية والسندات والأرباح المحتجزة.

13.2 Opportunity identification and market research

An **opportunity** exists when a group of customers has a problem or desire that is not served well, they are willing and able to pay for a better solution, and you can deliver it at a cost below what they will pay, in a way competitors cannot immediately copy.

Sources of opportunities:

- **Pain points observed at work.** Many strong B2B companies start when an employee sees a recurring problem (e.g., manual customs paperwork, slow supplier payments).
- **Changes in the environment.** New regulation (e-invoicing mandates), new technology (APIs, AI), demographic shifts, infrastructure (a new port or free zone), trade-policy changes.
- **Inefficient markets.** Fragmented supply, opaque prices, slow intermediaries.
- **Imitation with adaptation.** Bringing a proven model to a new geography with local adaptation — common and legitimate, but competition follows quickly.

Market research methods

Method	What it tells you	Cost	Weakness
Desk research (reports, statistics, trade data)	Market size, growth, players, prices	Low	Often outdated or too aggregated
Customer interviews (15–30)	Problems, current solutions, decision process, willingness to pay	Low	Small sample; people say what they think you want to hear
Surveys	Quantified preferences	Medium	Stated vs actual behaviour
Competitor analysis	Prices, positioning, weaknesses	Low	Hard to see their costs
Pilot / pre-sale	Real willingness to pay	Medium	Requires a product or credible promise
Trade data (import statistics by HS code)	Volumes and origins of goods	Low	Requires interpretation

Sizing the market: TAM, SAM, SOM

- **TAM (total addressable market):** total demand for the category if you had 100% share.
- **SAM (serviceable available market):** the part you can realistically serve with your product, channel and geography.
- **SOM (serviceable obtainable market):** the share you can plausibly win in 3–5 years.

> **WORKED EXAMPLE · Sizing a B2B market (fictional)**
>
> A founder plans software that automates customs declarations for freight forwarders in three Gulf countries.
>
> - Licensed forwarders in the three countries: ~2,400 (assumption from trade-directory research).
> - Target: forwarders filing more than 100 declarations a month: ~900 (SAM in accounts).
> - Expected price: $500 per month → SAM = 900 × $6,000 = **$5.4m per year**.
> - Realistic 5-year share: 15% → SOM ≈ 135 customers ≈ **$0.8m ARR**.
>
> Insight: the market is real but small. A venture-capital investor will ask how the business expands (more countries, more products such as payments or trade finance) — otherwise this is a good lifestyle or bootstrapped business, not a VC case.

13.3 Customer identification

A professional describes the customer precisely. In B2B, define the **ideal customer profile (ICP)**: industry, size, geography, technology used, trigger events (new regulation, expansion), and the **buying centre** — the people involved in a purchase.

Role in the buying centre	Typical title	What they care about
Economic buyer	CFO, CEO, owner	ROI, budget, risk
User	Operations staff	Ease, time saved
Technical evaluator	IT, engineering	Integration, security
Procurement	Purchasing manager	Price, terms, compliance
Champion	Often a manager who feels the pain	Personal success, recognition
Blocker	Anyone threatened by change	Status quo

In B2C, define segments by needs and behaviour, not only demographics. “Women 25–40” is a demographic; “working parents who order groceries online weekly and value delivery slots” is a usable segment.

13.4 From launch to scale: what changes

Dimension	Launch	Growth	Maturity
Founder role	Does everything	Hires managers, sets direction	Governs, allocates capital
Decisions	Fast, intuitive	Need data and processes	Formal approvals (DoA)
Systems	Spreadsheets, basic accounting	ERP, CRM, BI	Integrated systems, automation
Finance	Cash survival	Working-capital funding, controls	Capital structure, dividends
Risk	Existential	Operational and credit	Strategic and disruption
Culture	Informal	Strained by new hires	Bureaucracy risk

PROFESSIONAL INSIGHT · What experienced founders know

The most dangerous period is often not the start but the transition to growth. Volumes rise faster than systems, controls and working-capital funding. Typical symptoms: invoices not issued on time, stock counts that do not match the system, customers paying late because nobody chases, and the bank overdraft maxed out despite record sales.

13.5 Exit or closure

Owners realise value through: trade sale (to a strategic buyer), sale to private equity, management buyout (MBO), initial public offering (IPO), dividends over time, or orderly closure. Closure can be **solvent** (members' voluntary liquidation: assets cover all debts) or **insolvent** (creditors' processes under insolvency law). Chapter 19 and Part XII cover restructuring and insolvency.

PRACTICE · Practice questions

1. For a planned business importing solar water pumps to farmers in a North African country, list five research questions and the method you would use for each.
2. A founder estimates TAM as "the $40bn agricultural equipment market". Critique this and propose a better approach.
3. Identify the buying centre for a hospital purchasing a new inventory-management system.

ANSWER · Answers to the practice questions

1. Examples: (a) How many farms use diesel pumps, and of what size? — government agricultural census, interviews with dealers. (b) Diesel cost per farm per year? — farmer interviews, fuel price data. (c) Willingness and ability to pay; need for financing? — interviews, microfinance institutions. (d) Import duties, standards and subsidies? — customs tariff, ministry regulations, interviews with customs brokers. (e) Competing suppliers and prices? — dealer visits, online catalogues, trade data by HS code.
2. That number is irrelevant: it includes tractors and harvesters in all countries. Build bottom-up: number of target farms × share likely to adopt × average spend. Then SAM by reachable regions and channels; SOM by realistic share.
3. Economic buyer: hospital director/CFO; users: pharmacy and stores staff, nurses; technical evaluator: IT and information security; procurement: purchasing department; clinical stakeholders (patient-safety lead); possible blockers: staff comfortable with the old system, IT if integration load is heavy.

CHAPTER 14

Business Models, Value Propositions, Revenue Models and Pricing

LEARNING OBJECTIVES · In this chapter

How businesses decide what to offer, to whom, how to charge, and at what price — and how these choices determine margins, cash flow and risk.

14.1 The Business Model Canvas

The Business Model Canvas (Osterwalder and Pigneur) is a one-page description of a business in nine blocks. It is widely used in startups and corporate innovation.

Block	Question	Example: fictional B2B spare-parts distributor
Customer segments	Who do we serve?	Oil-field service contractors; industrial plants
Value proposition	What problem do we solve?	Genuine parts within 24 hours; technical advice
Channels	How do we reach and deliver?	Key account managers; e-catalogue; own vans
Customer relationships	What relationship do we build?	Dedicated account manager; framework agreements
Revenue streams	How do we earn money?	Margin on parts; emergency delivery fee; service contracts
Key resources	What assets are essential?	Inventory; exclusive brand agreements; warehouse; technical staff
Key activities	What must we do well?	Procurement, inventory planning, fast logistics
Key partners	Who helps us?	Manufacturers (principals), freight forwarders, banks
Cost structure	What are the main costs?	Inventory carrying cost, salaries, warehouse, freight

14.2 Value propositions that win

A strong value proposition is specific, quantified and relevant to the buyer's economics. Compare:

- Weak: "We provide high-quality logistics solutions."
- Strong: "We clear your containers within 48 hours of arrival or we pay the demurrage — saving you on average $1,100 per container."

Common sources of B2B value: lower total cost (not just lower price), reduced risk, faster delivery, reliability, compliance, cash-flow benefits (credit terms, consignment stock), revenue growth for the customer, convenience and integration.

PROFESSIONAL INSIGHT · Total cost, not price

Experienced buyers compare **total cost of ownership**: price + freight + duties + inventory holding + failures + downtime + administration. A seller with a higher price but lower total cost can win if it proves the numbers. This is how premium suppliers compete against cheap imports.

14.3 Revenue models

Revenue model	How money is earned	Examples	Cash-flow profile	Key metric
Product sale (transaction)	Price × units	Distributors, manufacturers	Depends on credit terms	Gross margin, volume
Subscription	Recurring fee per period	SaaS, maintenance contracts	Predictable, often prepaid	ARR, churn
Usage-based	Fee per unit of use	Cloud computing, payments, telecoms	Variable, recurring	Volume, take rate
Commission / take rate	% of transaction value	Marketplaces, brokers, agents	Asset-light	GMV × take rate
Licensing / royalty	Fee for using IP	Brands, patents, franchises	High margin	Licensee sales
Spread / interest	Difference between buy and sell rates	Banks, traders, FX dealers	Balance-sheet heavy	Net interest margin, spread
Advertising	Payment for audience attention	Media, platforms	Volume-driven	Users, CPM
Project / contract	Milestone payments	Construction, consulting	Lumpy; advance payments critical	Backlog, margin
Freemium	Free base, paid premium	Software, apps	Conversion-dependent	Free-to-paid conversion
Razor and blades	Cheap base product, profitable consumables	Printers, coffee machines	Long-tail recurring	Installed base

14.4 Pricing strategies

Pricing is the most powerful profit lever. For a business with a 10% operating margin, a 1% price increase with no volume loss raises operating profit by about 10%.

Strategy	Logic	When it works	Risk
Cost-plus	Cost × (1 + markup)	Commodity-like goods, government contracts	Ignores customer value and competition
Competitive (market) pricing	Match or position against competitors	Transparent markets	Price wars
Value-based	Price based on value to customer	Differentiated B2B products	Requires quantified value
Penetration	Low price to win share	Network effects, scale economies	May never raise prices
Skimming	High price at launch, then lower	Innovative products	Invites competitors
Tiered / good-better-best	Versions at different prices	Software, services	Complexity
Dynamic	Prices change with demand	Airlines, hotels, freight	Customer resentment
Bundling	Package priced below sum	Services + products	Hides product profitability
Price discrimination	Different prices for segments	Students, regions, volume tiers	Legal limits; arbitrage

WORKED EXAMPLE · Price increase vs volume loss

A product sells 10,000 units at $100; variable cost $70; fixed costs $200,000.

- Current profit = 10,000 × (100 − 70) − 200,000 = **$100,000**.
- Increase price 5% to $105: contribution per unit rises to $35. Break-even volume to keep the same profit: (100,000 + 200,000) ÷ 35 = **8,571 units** — the business can lose up to 14.3% of volume and still be as profitable.
- Cut price 5% to $95: contribution falls to $25. Required volume = 300,000 ÷ 25 = **12,000 units** (+20%).

Rule: with thin contribution margins, discounts require very large volume increases to pay off.

FORMULA · Volume change needed to offset a price change

Required volume change % = $-\Delta P \div (CM + \Delta P)$

where ΔP = price change as % of price, CM = contribution margin % before the change.

Example: CM = 30%, $\Delta P = -5\% \rightarrow -(-5) \div (30 - 5) = +20\%$ volume needed.

14.5 Discounts, terms and the hidden price

The **pocket price** is what the seller actually keeps after all deductions: invoice discounts, volume rebates, free freight, extended payment terms (financing cost), marketing allowances, returns. In many B2B businesses the pocket price is 10–20% below list price, and different customers have very different pocket prices without management noticing.

WORKED EXAMPLE · Pocket price waterfall (fictional)

Item	$/unit
List price	100.00
Invoice discount 8%	(8.00)
Volume rebate 3%	(2.76)
Free delivery	(2.10)
Payment terms 90 days vs standard 30 (60 extra days at 9% p.a. on $89.24)	(1.32)
Promotional allowance	(1.50)
Pocket price	**84.32**

14.6 Cost structures

Cost type	Definition	Examples	Management implication
Variable	Change with volume	Materials, freight, commissions	Determine contribution margin
Fixed	Do not change with volume in the short term	Rent, salaries, depreciation, insurance	Create operating leverage
Semi-variable	Fixed base + variable element	Utilities, overtime	Split for analysis
Step	Fixed within a range, then jump	Adding a shift, a second warehouse	Capacity planning
Direct	Traceable to a product	Raw materials, direct labour	Product costing
Indirect	Shared across products	Management, IT	Allocation choices matter

Cost type	Definition	Examples	Management implication
Sunk	Already spent, unrecoverable	Past R&D	Ignore in decisions
Opportunity	Value of best alternative forgone	Using warehouse space for product A instead of B	Include in decisions

Operating leverage measures how much operating profit changes for a change in revenue. Businesses with high fixed costs (airlines, software, telecoms, shipping) see profits swing sharply with volume.

FORMULA · Degree of operating leverage (DOL)

DOL = Contribution margin ÷ Operating profit = %Δ Operating profit ÷ %Δ Revenue

Example: contribution $3m, operating profit $1m → DOL = 3. A 10% revenue fall reduces operating profit by about 30%.

PRACTICE · Practice questions

1. Describe the revenue model and the key metric for: (a) a container shipping line; (b) a B2B payments platform; (c) a construction contractor; (d) a franchise brand owner.
2. A distributor has a 22% contribution margin. A customer asks for an 8% discount, promising more volume. How much more volume is needed to keep contribution unchanged?
3. A company's list price is $50. Typical deductions: 10% invoice discount, 2% rebate on net invoice, free shipping costing $1.20, 120-day terms instead of 30 (cost of capital 10%). Compute the pocket price.
4. Company X has fixed costs of $6m and variable costs of 40% of revenue; revenue is $12m. Compute operating profit and DOL. What happens to profit if revenue falls 15%?

ANSWER · Answers to the practice questions

1. (a) Freight per container/TEU (spot and contract); metric: rate per TEU, utilisation, unit cost. (b) Take rate on payment volume, sometimes subscription; metric: total payment volume × take rate, transaction margin. (c) Project/contract revenue recognised over time; metrics: backlog, gross margin per project, cash (advance payments vs work-in-progress). (d) Royalties (% of franchisee sales) + initial fees; metric: system-wide sales, number of units, royalty rate.
2. Required volume change = 8 ÷ (22 − 8) = **57.1%**. Very unlikely — the discount should be refused or exchanged for other value (longer contract, prepayment, exclusivity).
3. After 10%: 45.00. Rebate 2% of 45: −0.90 → 44.10. Shipping −1.20 → 42.90. Extra 90 days financing: 42.90 × 10% × 90/365 = 1.06 → pocket price ≈ **$41.84** (16.3% below list).
4. Contribution = 60% × 12 = $7.2m; operating profit = 7.2 − 6 = **$1.2m**; DOL = 7.2 ÷ 1.2 = **6**. Revenue −15% → contribution falls to 6.12 → profit **$0.12m** (−90%, consistent with DOL 6 × 15%).

CHAPTER 15

Cost Structures, Operations and the Functional Departments

LEARNING OBJECTIVES · In this chapter

What each department does, what it measures, what it needs from the others, and which documents and systems connect them.

15.1 The value chain of a typical company

Michael Porter's value chain divides activities into **primary activities** (inbound logistics, operations, outbound logistics, marketing and sales, service) and **support activities** (firm infrastructure — finance, legal, management; human resources; technology; procurement). Profit is the difference between the value customers pay for and the total cost of these activities.

15.2 The departments

Department	Core responsibilities	Key metrics	Key documents / systems	Depends on
Sales	Win and grow customers; negotiate prices and terms	Revenue, margin, win rate, pipeline	Quotations, contracts, CRM	Operations (delivery), finance (credit approval), marketing (leads)
Marketing	Demand generation, brand, pricing research	Leads, CAC, brand awareness	Campaign plans, analytics	Sales feedback, product
Procurement	Source suppliers, negotiate, buy	Purchase price variance, supplier OTIF, savings	RFQs, POs, supplier contracts	Demand forecasts (sales), specs (operations), payment terms (finance)
Operations / production	Make or deliver the product or service	Unit cost, capacity utilisation, quality, OTIF	Production plans, work orders	Materials (procurement), forecasts (sales)
Logistics / supply chain	Move and store goods	Lead time, inventory turnover, freight cost, fill rate	Shipping documents, WMS/TMS	Forwarders, customs, procurement
Customer service	Support, complaints, returns	Response time, NPS, resolution rate	Tickets, CRM	Operations, sales
Finance & accounting	Record, report, plan, fund, control	Cash, profit, DSO, budget variance	Ledgers, statements, budgets, ERP	Everyone's data
Treasury	Liquidity, banking, FX, funding	Cash position, funding cost	Cash forecasts, facility agreements	Finance, sales/procurement (timing)
Credit control	Customer credit limits and collections	DSO, overdue %, bad debt	Credit applications, ageing reports	Sales (customers), finance
Human resources	Hire, pay, develop, retain people	Turnover, time to hire, engagement	Contracts, payroll, policies	Budgets (finance), managers
Legal & compliance	Contracts, disputes, regulatory compliance	Disputes, compliance incidents	Contract templates, policies	Sales, procurement, management

Department	Core responsibilities	Key metrics	Key documents / systems	Depends on
IT	Systems, data, security	Uptime, incidents, project delivery	Architecture, SLAs	Business requirements
Risk / internal audit	Identify risks; independent assurance	Audit findings, risk indicators	Risk registers, audit reports	All departments

بالعربية — الإدارات ووظائفها

كل إدارة في الشركة تعتمد على غيرها: المبيعات تحتاج من العمليات أن تسلّم في الموعد، ومن المالية أن توافق على الحد الائتماني للعميل. والمشتريات تحتاج إلى توقعات المبيعات لتحديد الكميات، وإلى المالية لتحديد شروط الدفع. والمالية تعتمد على بيانات جميع الإدارات لإعداد التقارير والموازنات. فهم هذه العلاقات هو مفتاح فهم كيف تعمل الشركة فعلياً.

15.3 The order-to-cash and procure-to-pay cycles

Two end-to-end processes link most departments and drive working capital.

Order-to-cash (O2C)

1. Customer enquiry → quotation (sales).
2. Credit check and credit limit (credit control).
3. Order acceptance; sales order created in ERP.
4. Stock allocation or production order (operations).
5. Picking, packing, shipping; delivery note / proof of delivery (logistics).
6. Invoice issued (finance) — revenue recognised when control transfers.
7. Collection: reminders, dunning, cash application (credit control, treasury).
8. Reconciliation: payment matched to invoice; disputes resolved.

Procure-to-pay (P2P)

1. Need identified → purchase requisition.
2. Sourcing: RFQ/RFP, supplier selection (procurement).
3. Purchase order approved under the DoA.
4. Goods received / service confirmed: goods receipt note (GRN).
5. Supplier invoice received.
6. **Three-way match**: PO vs GRN vs invoice (quantity, price).
7. Payment scheduled per terms; payment executed (treasury) with approvals.
8. Accounting entries; supplier statement reconciliation.

COMMON MISTAKES · Common control failures in these cycles

- Invoices issued late or not at all after delivery (revenue leakage, longer DSO).
- Orders accepted for customers over their credit limit by sales under pressure.
- Payments made without three-way match — duplicate or fraudulent invoices paid.
- Supplier bank details changed on the strength of an email.
- Goods released before payment where terms required prepayment.

15.4 Operations: capacity, utilisation and quality

Operations converts inputs to outputs. The key concepts:

- **Capacity:** maximum output in a period. **Utilisation** = actual ÷ capacity. High utilisation lowers unit cost but increases waiting times and the risk of failure; most service operations become unstable above ~85–90% utilisation.
- **Throughput:** output per unit time; limited by the **bottleneck** (Theory of Constraints).
- **Quality:** cost of quality = prevention + appraisal + internal failure + external failure costs. Preventing defects is cheaper than fixing returns.
- **Make or buy:** produce internally or outsource, considering cost, control, quality, capacity and strategic importance.

WORKED EXAMPLE · Unit cost and utilisation (fictional bottling plant)

Fixed cost $1.2m per month; variable cost $0.10 per bottle; capacity 20m bottles/month.

- At 60% utilisation (12m bottles): unit cost = 0.10 + 1.2 ÷ 12 = **$0.200**.
- At 90% (18m): unit cost = 0.10 + 1.2 ÷ 18 = **$0.167** (−17%). A competitor running at 90% can undercut by 15% and still earn the same margin — this is why capacity utilisation drives pricing in manufacturing, shipping and airlines.

PRACTICE · Practice questions

1. Map the departments involved when a customer requests 60-day payment terms on a large order.
2. Explain the three-way match and the fraud it prevents.
3. A logistics warehouse runs at 95% utilisation. The CEO wants to accept a new customer that would add 10% volume. What would you advise?

ANSWER · Answers to the practice questions

1. Sales (negotiates), credit control (assesses limit and risk), finance/treasury (cost of funding the receivable, facility headroom), operations/logistics (capacity and delivery), legal (contract terms), possibly management (DoA approval for non-standard terms), and possibly credit insurance or the bank (receivables finance).
2. Matching PO (what was ordered, at what price), GRN (what was received) and invoice (what is billed) before payment. It prevents paying for goods not ordered or not received, duplicate invoices, price inflation and fictitious suppliers (when combined with vendor master controls).
3. 105% utilisation is impossible without extra capacity; even approaching 100% causes delays and errors for all customers. Options: add shifts or temporary space (step cost), price the new business to cover the incremental capacity, phase onboarding, or decline. Calculate the incremental contribution versus the cost of extra capacity and the risk to existing clients.

CHAPTER 16

How Departments Interact — and Why Profitable Businesses Fail

LEARNING OBJECTIVES · In this chapter

A single commercial decision traced through every department and all three financial statements; the mechanics of the working-capital trap; and the recurring reasons profitable or high-revenue companies get into financial distress.

16.1 The ripple effect of one sales decision

Gulf Kitchen Supplies LLC (fictional) distributes commercial kitchen equipment. The sales director wins a contract with a hotel group: **$6m of extra annual sales**, at a 10% price discount compared with list, with **120-day payment terms** instead of the standard 45. The board must decide whether to accept.

Baseline (before the contract)

Item	Value
Annual revenue	$20m
Gross margin	28%
Operating expenses (mostly fixed)	$3.6m
Operating profit	$2.0m
DSO	50 days
DIO (inventory days on COGS)	90 days
DPO	45 days
Overdraft limit / drawn	$4.0m / $2.5m
Interest rate on overdraft	9%

Effects across the company

Department / area	Effect of the new contract
Sales	+$6m revenue; commission paid on signing; account management workload.
Pricing / margin	List gross margin 28%; the 10% discount reduces gross margin on this contract to 20% (price 90, cost 72).
Procurement	Must buy ~$4.8m more stock per year; supplier wants 30% advance on larger orders.
Inventory / warehouse	Hotel requires 60 days of buffer stock dedicated to it; extra racking and a part-time storekeeper.
Logistics	Deliveries to 9 sites; installation teams scheduled.
Credit control	Single customer exposure rises to $2.0m at peak — above the credit-insurance limit of $1.2m.
Finance / working capital	Receivables, inventory and supplier advances all rise.
Treasury / banking	Overdraft headroom ($1.5m) is not enough; a facility increase or receivables finance is needed.
Legal	Hotel contract has liquidated damages of 1% per week of late delivery, capped at 10%.
HR	Two installers to hire.

Department / area	Effect of the new contract
Management	Concentration: hotel group would be 23% of revenue.

The numbers

Contribution from the contract:

- Revenue: $6.0m. COGS at 80% of discounted price: $4.8m. **Gross profit: $1.2m.**
- Incremental operating costs: installers $120k, storekeeper and racking $60k, extra delivery $90k, commissions 2% = $120k → **$390k**.
- Incremental operating profit **before financing: $810k**.

Working capital needed:

- Receivables: 6.0m × 120/365 = **$1.97m**.
- Dedicated inventory: 4.8m × 60/365 = **$0.79m** (plus normal pipeline stock 4.8 × 30/365 = $0.39m) → **$1.18m**.
- Supplier advance: 30% on, say, quarterly orders of $1.2m = **$0.36m** outstanding at peak.
- Less payables: 4.8m × 45/365 on the 70% not prepaid ≈ **($0.41m)**.
- **Net working capital required ≈ $3.1m.**

Financing cost at 9%: about **$280k per year**, and the existing overdraft headroom is only $1.5m.

Incremental profit after financing ≈ 810 − 280 = **$530k** before tax. Return on the incremental working capital: 810 ÷ 3,100 ≈ **26%** pre-financing — attractive *if* it can be funded and the customer pays.

> **PROFESSIONAL INSIGHT · How an experienced CFO would respond**
>
> "Commercially attractive, but not on these terms. We need (1) a facility increase or a receivables-finance line of at least $2m before signing; (2) credit insurance or a parent-company guarantee covering the exposure above $1.2m; (3) payment terms of 90 days rather than 120, or a 20% advance; (4) liquidated damages capped at 5%; (5) a price-review clause if supplier prices rise more than 5%."
>
> The graduate looks at the $6m revenue; the CFO looks at the $3.1m of cash it consumes and the $2m concentrated credit exposure.

Effect on the three statements (first year, simplified)

Statement	Line	Change
Income statement	Revenue	+6.0m
	COGS	−4.8m
	Operating expenses	−0.39m
	Interest	−0.28m
	Profit before tax	+0.53m
Balance sheet	Receivables	+1.97m
	Inventory	+1.18m
	Supplier advances	+0.36m
	Payables	+0.41m (liability)
	Borrowings	+2.6m (approx., after retained profit)
Cash flow	Operating cash flow	Profit 0.53m − ΔWC 3.1m ≈ **−2.57m**
	Financing	+2.6m borrowing

The contract makes the company **more profitable and more indebted** in its first year. If the hotel group pays late or disputes deliveries, profit evaporates and the debt remains.

16.2 The working-capital trap

The **cash conversion cycle** (CCC) measures how long cash is tied up in operations:

> **FORMULA · Cash conversion cycle**
>
> CCC = DIO + DSO − DPO
>
> DIO = inventory ÷ COGS × 365 · DSO = receivables ÷ revenue × 365 · DPO = payables ÷ COGS × 365 (purchases can be used instead of COGS where available)
>
> Working-capital need ≈ daily COGS × DIO + daily revenue × DSO − daily COGS × DPO

Every additional day of CCC on $20m revenue costs about $55k of cash (20m ÷ 365). Fast growth multiplies this. A company growing revenue at 40% a year with a 100-day CCC needs to finance roughly 0.4 × revenue × 100/365 ≈ 11% of current revenue in new working capital every year — usually more than its net margin can fund.

> **FORMULA · Self-funded growth rate (approximation)**
>
> Maximum growth funded by internal cash ≈ Net margin (after dividends) ÷ (working capital as % of sales + net fixed assets as % of sales − debt capacity created)
>
> A simplified version for working-capital-heavy traders: g ≈ retained net margin ÷ (NWC ÷ Sales). Example: retained margin 4%, NWC/Sales 25% → g ≈ 16% per year without new debt or equity.

16.3 Why profitable businesses fail

Cause	Mechanism	Warning signs	Typical remedy
Overtrading	Growth outruns working-capital funding	Rising DSO/DIO, overdraft at limit, late supplier payments	Slow growth, raise funding, tighten terms
Customer concentration	One customer delays or fails	Top customer > 20–25% of revenue	Credit insurance, diversification
Short-term funding of long-term assets	Capex funded with overdraft; bank withdraws line	Current ratio < 1, overdraft hardcore	Term loan, equity
Debt maturity wall	Loan matures when markets are closed	Large bullet repayments	Refinance early, stagger maturities
Covenant breach	Bank can accelerate	Shrinking headroom	Early negotiation, waivers
FX mismatch	Revenue in local currency, debt in USD; devaluation	Unhedged foreign debt	Hedging, matching
Fixed-price contracts in inflation	Costs rise, prices fixed	Long contracts without indexation	Escalation clauses
Fraud or control failure	Cash leaks, false profits	Unreconciled accounts, dominant individual	Controls, audit
Tax or legal shock	Large unexpected liability	Aggressive tax positions, disputes	Provisions, settlements
Supplier failure / disruption	Cannot deliver; penalties	Single-source supply	Dual sourcing, inventory buffers
Accounting profit not real	Revenue recognised too early; capitalised costs	Profit rising, cash falling	Quality-of-earnings review

بالعربية — لماذا تتعثر الشركات الرابحة؟

السبب الأكثر شيوعاً هو «الإفراط في المتاجرة»: تنمو المبيعات أسرع من قدرة الشركة على تمويل رأس المال العامل، فتتراكم الذمم المدينة والمخزون، ويُستنفد الحساب الجاري المدين، وتتأخر الشركة في الدفع للموردين، ثم يفقد البنك الثقة. ومن الأسباب الأخرى: تركّز الإيرادات في عميل واحد، وتمويل الأصول طويلة الأجل بقروض قصيرة الأجل، وعدم تطابق العملات، والعقود ذات الأسعار الثابتة في فترات التضخم، والاحتيال وضعف الرقابة الداخلية.

16.4 High revenue, financial distress: three patterns

1. **Low-margin trading with high volume.** A commodity trader with $500m revenue and a 1% net margin earns $5m. One defaulted cargo of $15m wipes out three years of profit. Revenue says nothing about resilience; equity and risk controls do.
2. **Project businesses with negative cash flows.** A contractor with $300m of revenue recognised over time may have $90m of unbilled work-in-progress and retentions, while paying subcontractors monthly. A client's payment delay pushes it into distress.
3. **Subsidised growth.** A delivery platform with $200m of gross merchandise value and negative unit economics loses more as it grows; it survives only while investors fund it.

PRACTICE · Practice questions

1. A company has revenue $36.5m, COGS $29.2m, receivables $6.0m, inventory $4.8m and payables $3.2m. Compute DSO, DIO, DPO and CCC. What is the cash impact of reducing DSO by 10 days?
2. Revisit the Gulf Kitchen case. The hotel offers 90-day terms if you remove the discount entirely. Recompute the gross profit and receivables. Which deal is better?
3. List five questions a bank credit officer would ask Gulf Kitchen when it requests a $2m facility increase.

ANSWER · Answers to the practice questions

1. DSO = 6.0 ÷ 36.5 × 365 = **60 days**; DIO = 4.8 ÷ 29.2 × 365 = **60 days**; DPO = 3.2 ÷ 29.2 × 365 = **40 days**; CCC = **80 days**. Reducing DSO by 10 days releases 36.5m ÷ 365 × 10 = **$1.0m** of cash.
2. Without discount the price is list: revenue for the same volume = 6.0 ÷ 0.9 = $6.67m; COGS $4.8m; gross profit **$1.87m** (vs $1.2m). Receivables = 6.67 × 90/365 = **$1.64m** (vs $1.97m). Clearly better: + $0.67m gross profit and $0.33m less receivables — if the hotel accepts it.
3. What is the hotel group's credit quality and payment history? How concentrated will exposure be, and is it insured? What is the working-capital forecast month by month? What are the contract's penalty and termination terms? What security and covenants can the company offer, and what is current leverage and headroom? (Also: management capacity to deliver, supplier terms.)

CHAPTER 17

Industry Economics: How Eight Industries Make Money

LEARNING OBJECTIVES · In this chapter

For each industry: the revenue model, cost structure, capital intensity, working-capital profile, key metrics, typical risks and how banks finance it. Figures are illustrative ranges for teaching and vary widely by company, country and period.

17.1 Comparative overview

Industry	Revenue model	Cost structure	Working capital	Key metrics	Typical financing
Manufacturing	Product sales	High fixed (plant), material-heavy	Inventory-heavy (raw, WIP, finished)	Utilisation, unit cost, yield, OEE	Term loans for plant; working-capital lines
Retail	Product sales to consumers	Rent, staff, inventory	Cash sales; supplier credit can make CCC negative	Same-store sales, sales per m², GMROI, shrinkage	Supplier credit, RCF, leases
Logistics	Freight, handling, storage fees	Fleet/warehouse fixed costs or asset-light	Receivables 45–90 days; pays carriers faster	Revenue per shipment, yield, utilisation	Asset finance, invoice finance
Software (SaaS)	Subscriptions	High R&D and sales; low cost to serve	Often negative (annual prepayments)	ARR, NRR, CAC payback, Rule of 40	Equity (VC), recurring-revenue loans
Banking	Interest spread + fees	Funding cost, staff, credit losses, capital	Balance sheet is the product	NIM, cost-to-income, NPL ratio, CET1, ROE	Deposits, wholesale funding, capital
Construction	Contracts recognised over time	Subcontractors, materials, plant	Retentions, WIP, advance payments	Backlog, margin per project, cash curve	Advance-payment and performance guarantees, project lines
Energy	Commodity sales, regulated tariffs, PPAs	Very capital-intensive	Varies	Production cost/barrel, lifting cost, capacity factor	Project finance, reserve-based lending, bonds
International trading	Margin on buy-sell	Freight, insurance, finance cost	Transaction-by-transaction	Margin per tonne, turnover of capital, VaR	Trade finance (L/Cs, borrowing base)

17.2 Manufacturing

A manufacturer buys raw materials, converts them with labour and machines, and sells finished goods. Profitability depends on **capacity utilisation**, **yield** (good output ÷ input), **material price** management and **product mix**. Overall equipment effectiveness (OEE = availability × performance × quality) is a standard operational KPI.

Cash cycle: raw materials (30–90 days) + production (days to months) + finished goods (30–60 days) + receivables (45–90 days) − payables (30–60 days). CCC of 90–150 days is common.

WORKED EXAMPLE · Fictional plastic-packaging plant

> Revenue $40m; materials 55% of revenue (polymer resin priced off oil derivatives); direct labour 8%; energy 6%; plant overheads and depreciation $6m; SG&A $3m. EBITDA margin ≈ 18%, EBIT ≈ 13%. A 20% rise in resin prices not passed through for one quarter cuts quarterly EBIT by about 85% (55% × 20% = 11% of revenue). Mitigation: price-adjustment formulas linked to resin indices.

17.3 Retail

Retailers earn a gross margin (typically higher in fashion, lower in grocery) and must cover rent, staff and logistics. Strong grocers can have a **negative cash conversion cycle**: they sell stock for cash in days but pay suppliers in 45–60 days, so suppliers finance their growth. Key metrics: like-for-like (same-store) sales growth, sales per square metre, basket size, footfall, conversion, inventory turnover, shrinkage (losses from theft and damage), GMROI (gross margin ÷ average inventory at cost). E-commerce adds fulfilment and return costs and customer-acquisition spending.

17.4 Logistics

Two models: **asset-heavy** (own trucks, warehouses, vessels — high fixed costs, utilisation-driven) and **asset-light** (freight forwarders, brokers, 3PL management — buy capacity from carriers, earn a margin or fee). Forwarders report **gross profit per shipment/TEU** rather than revenue because pass-through freight inflates revenue when freight rates spike. A forwarder's revenue can double in a freight-rate boom with almost no change in gross profit — another reason not to judge businesses by revenue.

Working capital risk: forwarders often pay carriers within 7–30 days but collect from customers in 45–90 days.

17.5 Software (SaaS)

Gross margins of 70–85% because serving an extra customer costs little; heavy upfront spending on development and customer acquisition. Revenue is recognised monthly while customers often pay annually in advance, creating deferred revenue and favourable cash flow. Metrics: ARR, net revenue retention, gross churn, CAC payback months, LTV/CAC, and the "Rule of 40" (growth % + EBITDA or FCF margin % ≥ 40 as a heuristic for balance between growth and profitability).

17.6 Banking

A bank's products are its balance sheet. It borrows (deposits, wholesale funding) and lends at a higher rate, earning **net interest income**; it also earns fees (payments, trade finance, advisory). Costs: staff, technology, credit losses and the cost of holding regulatory capital. Key ratios: net interest margin (NIM), cost-to-income ratio, NPL ratio and coverage, capital ratios (CET1), liquidity ratios (LCR, NSFR), return on equity. Banks are highly leveraged (assets often 10–12× equity), so small losses on assets matter. Part VI explains banking in depth.

17.7 Construction

Contractors win projects through tenders, often providing **bid bonds**, then **performance guarantees** and **advance-payment guarantees**. Revenue is recognised over time as work progresses (IFRS 15 input or output methods). Clients often hold back **retentions** (5–10%) until completion and the defects liability period ends. Typical risks: underpriced bids, scope changes not approved (variations), delays with liquidated damages, subcontractor failure, material price inflation, and client payment delays. Cash curve: positive if advance payments and timely certifications exceed spending; negative if certifications lag.

17.8 Energy

Upstream oil and gas: exploration and production with very high capex, long lead times, commodity price exposure; financed by equity, bonds, reserve-based lending. Midstream: pipelines and storage, often contracted fee income. Downstream: refining (earns the **crack spread**) and marketing. Power: generation under long-term power purchase agreements (PPAs) — typical for renewable project finance — or merchant exposure. Utilities: regulated returns on a regulated asset base. Part XI covers commodity trading.

17.9 International trading

Traders buy from producers and sell to buyers across borders, earning a margin from price differences across locations, times, qualities and from logistics and credit services. Margins are thin (often 1–5% gross on commodities, more on specialised goods); return on capital depends on **turning capital quickly** and leveraging trade finance. Risks: counterparty default, price moves between purchase and sale, quality/quantity disputes, shipping problems, documentary discrepancies, sanctions and fraud.

WORKED EXAMPLE · Return on capital for a fictional trader

A trader buys $2m of goods and sells for $2.06m (3% gross margin) every 60 days, six times a year. Costs per deal: freight/insurance included in margin; finance cost (80% L/C-financed at 8% for 60 days) ≈ $21k; other costs $9k. Net per deal ≈ $30k; per year ≈ $180k. Equity employed: 20% × $2m = $400k → return on equity ≈ **45%**. One deal lost through default (−$2m) would wipe out five equity bases — so credit risk controls dominate trading.

PRACTICE · Practice questions

1. Why might a freight forwarder's revenue fall 40% while its gross profit stays stable?
2. Explain how a supermarket can have negative working capital, and why that is valuable.
3. A SaaS company grows 35% with an EBITDA margin of −10%. Does it pass the Rule of 40? What would you ask?
4. A contractor recognises $50m of revenue but has negative operating cash flow. Give three causes.

ANSWER · Answers to the practice questions

1. Revenue includes pass-through freight; when market freight rates fall, revenue falls while the forwarder's margin per shipment (gross profit) can remain stable.
2. It collects cash from consumers immediately (DSO ≈ 0), turns inventory quickly (e.g., 20 days) and pays suppliers in 45–60 days, so CCC is negative: suppliers finance its operations and growth generates cash.
3. 35 + (−10) = 25 < 40 → fails the heuristic. Ask about CAC payback, net revenue retention, gross margin, cash runway, and when operating leverage will lift margins.
4. Retentions withheld by the client; certifications/billing lagging work done (contract assets/WIP); client payment delays; heavy upfront procurement; unapproved variations.

CHAPTER 18

Growth, Foreign Expansion, M&A, Restructuring and Closure

LEARNING OBJECTIVES · In this chapter

How companies grow and how they shrink: organic and inorganic growth, entering foreign markets, mergers and acquisitions (overview), restructuring, insolvency and closure. Corporate-finance aspects of M&A are in Chapter 34 (Volume 3); legal aspects in Part XII; integration in Part XIV.

18.1 Growth strategies

The Ansoff matrix organises growth options:

	Existing products	New products
Existing markets	Market penetration (sell more to current customers/segments) — lowest risk	Product development
New markets	Market development (new regions, segments)	Diversification — highest risk

Organic growth (building yourself) is slower but cheaper and easier to control. **Inorganic growth** (acquisitions, joint ventures) is faster but riskier: overpayment, integration failure, culture clash.

18.2 Expansion into foreign markets

Entry mode	Control	Investment	Risk	Typical use
Exporting via agents/distributors	Low	Low	Low	Testing a market
Licensing / franchising	Low-medium	Low	IP and brand risk	Consumer brands, technology
Representative office	Low	Low	Limited activities permitted	Market intelligence, relationship building
Branch	High	Medium	Parent liable for branch	Banks, contractors (where permitted)
Joint venture	Shared	Medium	Partner disputes	Markets requiring local partners or knowledge
Wholly owned subsidiary	Full	High	Full exposure	Long-term commitment
Acquisition	Full	High	Integration, valuation	Fast access to customers and licences

Questions to answer before entering a country: market size and growth; regulatory restrictions on foreign ownership; licences; tax (corporate tax, withholding tax on dividends and royalties, double tax treaties, transfer pricing); currency convertibility and repatriation of profits; labour law; contract enforcement and dispute resolution; political and sanctions risk; payment practices; local partners' reputation (due diligence).

COMMON MISTAKES · Common mistakes in foreign expansion

- Assuming the home-country business model works unchanged.

- Choosing a local partner on personal trust without due diligence; giving exclusivity without performance targets.
- Ignoring how profits will be repatriated (withholding taxes, FX controls).
- Underestimating time to obtain licences and hire staff.
- Pricing in local currency without hedging or cost matching.

18.3 Mergers and acquisitions: an overview

Motives: market share, scale economies, new capabilities or technology, new geographies, vertical integration, tax or financial engineering (less defensible), and sometimes managerial empire-building. The typical process:

1. Strategy and target screening.
2. Approach; NDA; preliminary information (teaser, information memorandum).
3. Non-binding offer / letter of intent; valuation range.
4. Due diligence (financial, tax, legal, commercial, operational, IT, HR, ESG).
5. Binding offer; negotiation of the sale and purchase agreement (SPA).
6. Signing; conditions precedent (regulatory approvals, financing).
7. Closing; payment; completion accounts or locked box.
8. Post-merger integration.

Studies and practitioner experience consistently show that many acquisitions fail to create value for the acquirer, mainly through overpayment and integration problems.

18.4 Restructuring

Restructuring changes a company's operations, capital structure or ownership to restore viability or increase value.

Type	What changes	Examples
Operational	Costs, footprint, processes	Closing branches, outsourcing, headcount reduction
Financial	Debt and equity	Maturity extension, covenant reset, debt-for-equity swap, new money
Portfolio	Businesses owned	Divestments, spin-offs
Legal/organisational	Group structure	Merging entities, holding company changes

A financial restructuring usually involves negotiations with lenders under a **standstill** (lenders agree not to enforce while a plan is prepared), an **independent business review**, a **13-week cash-flow forecast**, and a plan that shares the losses among shareholders and creditors according to their priority.

18.5 Bankruptcy, insolvency and closure

Insolvency is generally tested in two ways: **cash-flow insolvency** (unable to pay debts as they fall due) and **balance-sheet insolvency** (liabilities exceed assets). Each jurisdiction has its own procedures — for example reorganisation procedures that allow the business to continue under court protection while a plan is agreed (US Chapter 11 is the best-known example), administration-type procedures, preventive composition or restructuring procedures, and liquidation. Several Middle Eastern countries have modernised their bankruptcy laws in recent years; the details must be verified locally with a qualified lawyer.

The **priority of claims** (waterfall) in liquidation typically runs: secured creditors (from their collateral) → insolvency costs → preferential creditors (often certain employee claims and taxes, depending on the law) → unsecured creditors → subordinated creditors → shareholders. Shareholders usually receive nothing in an insolvent liquidation.

بالعربية — الإعسار والإفلاس

يختلف قانون الإفلاس من بلد إلى آخر، لكن الفكرة العامة واحدة: عندما تعجز الشركة عن سداد ديونها في مواعيدها، أو تزيد التزاماتها على أصولها، يمكن اللجوء إلى إجراءات قانونية لإعادة الهيكلة تحت حماية المحكمة أو إلى التصفية. ترتيب الأولويات في التوزيع يبدأ عادةً بالدائنين المضمونين ثم تكاليف الإجراءات ثم الديون الممتازة كبعض مستحقات العمال والضرائب ثم الدائنين العاديين، ويأتي المساهمون في المرتبة الأخيرة. يجب استشارة محامٍ مختص في كل دولة لأن التفاصيل تختلف كثيراً.

18.6 Directors' responsibilities near insolvency

In many jurisdictions, when a company is insolvent or close to it, directors must consider creditors' interests and may face personal liability if they continue trading irresponsibly, prefer certain creditors, or dispose of assets at undervalue. Practical steps: take professional advice early, keep accurate records, minute decisions, avoid new debts the company cannot pay, and communicate with lenders.

PRACTICE · Practice questions

1. A Lebanese software company wants to sell into Saudi Arabia. Compare exporting through a reseller with establishing a subsidiary.
2. A company has assets that would realise $30m in liquidation: $20m from a mortgaged warehouse (bank owed $25m) and $10m from other assets. Unsecured creditors are owed $15m; liquidation costs $1m; preferential claims $2m. Estimate recoveries for the bank and unsecured creditors.
3. Why do lenders usually prefer a consensual restructuring to liquidation?

ANSWER · Answers to the practice questions

1. Reseller: fast, low cost, low control, reseller margin 20–40%, dependent on reseller's motivation, limited customer data; check exclusivity and termination terms. Subsidiary: higher cost and time (licensing, local hiring, possible localisation requirements), more control and credibility with large/government clients, local tax obligations, repatriation planning. Many firms start with a reseller and move to a subsidiary once revenue justifies it.
2. Bank recovers $20m from its collateral; its $5m shortfall becomes an unsecured claim. Remaining $10m: costs $1m and preferential $2m → $7m for unsecured creditors totalling 15 + 5 = $20m → **35 cents per dollar**. Bank total ≈ 20 + 1.75 = **$21.75m (87%)**; other unsecured creditors receive **$5.25m (35%)**. (Order of costs vs secured realisations varies by jurisdiction.)
3. Liquidation destroys going-concern value (customers, staff, contracts), sells assets at distressed prices and incurs heavy costs; a viable business restructured usually returns more to creditors, faster.

PART III

Accounting and Financial Statements

المحاسبة والقوائم المالية

Accounting is the measurement system of business. A finance graduate knows the statements; the aim of this part is fluency: being able to record any transaction, see instantly how it moves the three statements, judge whether reported numbers are credible, and use accounting information for management decisions.

The part follows the accounting cycle: principles and bookkeeping mechanics (Chapter 19), recognition rules (Chapter 20), specific assets and liabilities (Chapter 21), the statements themselves with a complete worked example (Chapter 22), group accounts and standards (Chapter 23), management accounting (Chapter 24), analysis (Chapter 25) and controls and audit (Chapter 26).

LEARNING OBJECTIVES · **Learning objectives for Part III**

- Record transactions with correct journal entries and produce a trial balance and financial statements.
- Apply accrual concepts, revenue recognition (IFRS 15) and expense recognition.
- Account for inventory, receivables, fixed assets, intangibles, provisions and leases.
- Show how any single transaction affects the income statement, balance sheet and cash flow statement.
- Use budgets, cost accounting and variance analysis for management decisions.
- Analyse financial statements with ratios and judge earnings quality.
- Understand internal controls and what an audit does — and does not — provide.

STANDARD / FRAMEWORK · **Which standards are referred to**

IFRS Accounting Standards are issued by the International Accounting Standards Board (IASB) of the IFRS Foundation; older standards are called IAS. They are required or permitted for listed companies in many jurisdictions, including the EU, the UK, most of the Middle East and much of Asia; in several Gulf countries IFRS applies broadly to companies, but local requirements must be checked. **US GAAP** is set by the Financial Accounting Standards Board (FASB) and codified in the Accounting Standards Codification (ASC). Small entities may use simplified frameworks (e.g., IFRS for SMEs). This part explains principles at a level suitable for business professionals; accountants preparing statements must use the full current standards.

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CHAPTER 19

Accounting Foundations: Principles, Double Entry, Ledgers and Trial Balance

LEARNING OBJECTIVES · In this chapter

The concepts underlying financial statements, the rules of debit and credit, the chart of accounts, journal entries, the general ledger and the trial balance.

19.1 The purpose and qualities of financial information

The IASB's Conceptual Framework says the objective of general-purpose financial reporting is to provide information useful to existing and potential investors, lenders and other creditors in making decisions about providing resources to the entity.

Quality	Meaning	Practical implication
Relevance	Capable of making a difference to decisions (predictive and confirmatory value; materiality)	Report what matters, in enough detail
Faithful representation	Complete, neutral, free from error; substance over legal form	Do not hide liabilities in clever structures
Comparability	Users can compare across periods and entities	Consistent policies; disclose changes
Verifiability	Independent observers could reach consensus	Evidence and audit trails
Timeliness	Available in time to influence decisions	Fast monthly close
Understandability	Classified, characterised and presented clearly	Clear notes and segment information

Key underlying concepts

- **Going concern:** statements assume the entity will continue operating for the foreseeable future. If not, a different basis applies and assets may be worth far less.
- **Accrual basis:** effects of transactions are recorded when they occur, not when cash moves.
- **Entity concept:** the business is separate from its owners — the owner's personal expenses are not company expenses.
- **Prudence (caution):** exercising caution under uncertainty, without deliberately understating assets or overstating liabilities.
- **Substance over form:** economic reality prevails over legal form (e.g., a "sale" with an obligation to repurchase may in substance be a loan).
- **Materiality:** information is material if omitting or misstating it could reasonably influence users' decisions.
- **Measurement bases:** historical cost, fair value, current value; most operating assets use historical cost less depreciation and impairment; many financial instruments use fair value or amortised cost.

19.2 The accounting equation and the rules of debit and credit

FORMULA · The accounting equation

Assets = Liabilities + Equity

Equity = Share capital + Retained earnings (+ other reserves)
Retained earnings increase with revenue and decrease with expenses and dividends, so the expanded form is:
Assets + Expenses + Dividends = Liabilities + Share capital + Retained earnings (opening) + Revenue

Every transaction keeps the equation in balance. **Debit** (Dr) means the left side of an account; **credit** (Cr) the right side.

Account type	Increase	Decrease	Normal balance
Assets	Debit	Credit	Debit
Expenses	Debit	Credit	Debit
Dividends / drawings	Debit	Credit	Debit
Liabilities	Credit	Debit	Credit
Equity (capital, retained earnings)	Credit	Debit	Credit
Revenue / income	Credit	Debit	Credit

PROFESSIONAL INSIGHT · A memory aid that actually works

Think about **cash** first. Receiving cash is always a debit to cash (an asset increases). Paying cash is always a credit to cash. Then ask what the other side is: if you received cash because you sold something, credit revenue; because you borrowed, credit loans; because a customer paid an invoice, credit receivables. Bank statements confuse beginners because they are written from the bank's point of view: your deposit is the bank's liability, so the bank "credits" your account.

بالعربية — المدين والدائن

كلمتا «مدين» و«دائن» لا تعنيان «جيد» و«سيئ»، بل هما جانبا الحساب: المدين هو الجانب الأيسر، والدائن هو الجانب الأيمن. الأصول والمصاريف تزيد بالمدين، أما الالتزامات وحقوق الملكية والإيرادات فتزيد بالدائن. ولكل عملية طرفان على الأقل، ويجب أن يتساوى مجموع المدين مع مجموع الدائن. أسهل طريقة للتعلّم هي البدء بالنقد: استلام النقد يكون دائماً مديناً في حساب النقدية، ودفعه يكون دائناً.

19.3 The chart of accounts

A chart of accounts (CoA) lists all accounts with codes, grouped to produce the statements and management reports. A typical numbering:

Range	Group	Examples
1000–1999	Assets	1010 Cash at bank – USD; 1100 Trade receivables; 1200 Inventory; 1300 Prepayments; 1500 Property, plant and equipment; 1590 Accumulated depreciation; 1600 Intangible assets
2000–2999	Liabilities	2010 Trade payables; 2100 Accrued expenses; 2200 VAT payable; 2300 Deferred revenue; 2500 Bank loans; 2600 Lease liabilities; 2700 Provisions
3000–3999	Equity	3010 Share capital; 3100 Statutory reserve; 3200 Retained earnings

Range	Group	Examples
4000–4999	Revenue	4010 Sales – products; 4020 Sales – services; 4900 Other income
5000–5999	Cost of sales	5010 Cost of goods sold; 5020 Inbound freight; 5030 Customs duties
6000–7999	Operating expenses	6010 Salaries; 6100 Rent; 6200 Marketing; 6500 Depreciation; 6600 IT
8000–8999	Finance and tax	8010 Interest expense; 8100 FX gains/losses; 8900 Income tax

Good CoA design matters for a CS-minded professional: it is a data model. Combine the account code with **dimensions** (cost centre, project, product line, region, customer) rather than creating thousands of accounts. This enables multi-dimensional reporting in BI tools without duplicating the ledger.

19.4 Journal entries

A journal entry records the date, accounts, debit and credit amounts and a description, with supporting evidence (invoice, contract, bank advice).

WORKED EXAMPLE · Ten journal entries for a new trading company (fictional, USD)

#	Transaction	Debit	Credit	Amount
1	Owners inject capital	Cash at bank	Share capital	200,000
2	Bank loan received	Cash at bank	Bank loan	100,000
3	Buy delivery van for cash	Vehicles (PP&E)	Cash at bank	45,000
4	Buy inventory on credit	Inventory	Trade payables	120,000
5	Pay import duty and freight on that inventory	Inventory	Cash at bank	9,000
6	Sell goods on credit (selling price)	Trade receivables	Sales revenue	150,000
6b	Cost of the goods sold	Cost of goods sold	Inventory	96,750
7	Pay supplier part	Trade payables	Cash at bank	80,000
8	Customer pays	Cash at bank	Trade receivables	110,000
9	Pay salaries	Salaries expense	Cash at bank	18,000
10	Month-end: rent for the month not yet invoiced	Rent expense	Accrued expenses	6,000

Entry 6b: the goods sold were 75% of the inventory purchased (120,000 + 9,000 = 129,000 × 75% = 96,750). Inventory cost includes freight and duty (IAS 2 includes costs of purchase, including import duties and transport, in the cost of inventories).

19.5 The general ledger and T-accounts

Each journal entry is posted to ledger accounts. A **T-account** shows debits on the left, credits on the right.

```
            Cash at bank (1010)
Dr                                  Cr
(1) Capital       200,000  | (3) Van            45,000
(2) Loan          100,000  | (5) Duty/freight    9,000
(8) Customer      110,000  | (7) Supplier       80,000
                           | (9) Salaries       18,000
--------------------------+-----------------------------
Total             410,000  | Total             152,000
Balance c/d (Dr)  258,000
```

19.6 The trial balance

A trial balance lists all ledger balances. Total debits must equal total credits. It does **not** prove the accounts are correct: an entry posted to the wrong account, or omitted entirely, still balances.

WORKED EXAMPLE · Trial balance after the ten entries

Account	Debit	Credit
Cash at bank	258,000	
Trade receivables	40,000	
Inventory (129,000 − 96,750)	32,250	
Vehicles	45,000	
Trade payables (120,000 − 80,000)		40,000
Accrued expenses		6,000
Bank loan		100,000
Share capital		200,000
Sales revenue		150,000
Cost of goods sold	96,750	
Salaries expense	18,000	
Rent expense	6,000	
Totals	**496,000**	**496,000**

Profit for the month (before depreciation and interest): 150,000 − 96,750 − 18,000 − 6,000 = **29,250**. Chapter 22 adds depreciation, interest and tax and produces full statements.

19.7 The accounting cycle and the month-end close

1. Capture source documents (invoices, receipts, contracts, bank data).
2. Record journal entries (mostly automatically from sub-ledgers: sales, purchases, payroll, fixed assets).
3. Post to the general ledger.
4. Reconcile: bank, receivables and payables control accounts vs sub-ledgers, intercompany, inventory counts, VAT.
5. Adjusting entries: accruals, prepayments, depreciation, provisions, FX revaluation, inventory adjustments, deferred revenue.
6. Adjusted trial balance.
7. Financial statements and management reports.
8. Closing entries at year-end: revenue and expense accounts are closed to retained earnings.

Leading finance teams close the month in 3–5 working days, using automation for bank reconciliation, intercompany matching and accrual calculations.

COMMON MISTAKES · Common bookkeeping mistakes

- Recording personal or owner expenses in the company books.
- Treating loan principal repayments as expenses (only interest is an expense).
- Expensing capital items (a $40k machine) or capitalising repairs.
- Forgetting accruals: costs of the month invoiced next month.
- Recording customer advances as revenue.

- Not reconciling the bank account monthly.

PRACTICE · Practice questions

Record the journal entries:

1. A company issues a sales invoice of $11,500 including 15% VAT, payable in 30 days.
2. It pays a $24,000 annual insurance premium on 1 January for the calendar year.
3. Month-end entry for insurance at 31 January.
4. It repays $10,000 of a bank loan plus $750 interest.
5. A customer pays a $5,000 advance for goods to be delivered next month.
6. A trial balance balances. Does that mean the accounts are correct? Give two errors it would not detect.

ANSWER · Answers to the practice questions

1. Dr Trade receivables 11,500 / Cr Sales revenue 10,000 / Cr VAT payable 1,500. (VAT collected for the government is a liability, not revenue.)
2. Dr Prepayments 24,000 / Cr Cash 24,000.
3. Dr Insurance expense 2,000 / Cr Prepayments 2,000.
4. Dr Bank loan 10,000 / Dr Interest expense 750 / Cr Cash 10,750.
5. Dr Cash 5,000 / Cr Deferred revenue (contract liability) 5,000. Revenue is recognised next month on delivery.
6. No. It would not detect: an omitted transaction; a posting to the wrong account of the same type (e.g., rent posted to salaries); a compensating error; an entry with the wrong amount on both sides; a reversed entry (debit and credit swapped).

CHAPTER 20

Accrual Accounting, Revenue and Expense Recognition

LEARNING OBJECTIVES · In this chapter

When revenue and expenses are recognised; IFRS 15's five-step model with commercial examples; principal vs agent; how Incoterms and contract terms affect revenue timing; accruals and cut-off.

20.1 Cash accounting vs accrual accounting

Event	Cash accounting	Accrual accounting
Goods delivered in March, customer pays in May	Revenue in May	Revenue in March; receivable until May
Annual software licence paid in January	Expense in January	Prepayment, expensed monthly
Electricity used in December, billed in January	Expense in January	Accrued expense in December
Customer advance received in June for September delivery	Revenue in June	Contract liability until September
Machine bought for cash	Expense when paid	Asset; depreciation over useful life

Cash accounting is simple but misleading about performance; accrual accounting reflects performance but requires judgement and can be manipulated. Professionals use both views together: the income statement (accrual) and the cash flow statement (cash).

20.2 Revenue recognition: IFRS 15 and ASC 606

IFRS 15 *Revenue from Contracts with Customers* and the converged US standard ASC 606 use a five-step model:

Step	Requirement	Commercial meaning
1. Identify the contract	Enforceable agreement, approved, rights and payment terms identifiable, commercial substance, collection probable	No contract (or no probable collection) → no revenue
2. Identify performance obligations	Distinct goods or services promised	A sale of equipment + installation + 2-year maintenance may be 3 obligations
3. Determine the transaction price	Consideration expected, including variable consideration (rebates, penalties, bonuses) constrained to amounts highly probable not to reverse; significant financing components	Volume rebates and late-delivery penalties reduce revenue
4. Allocate the price	Based on relative standalone selling prices	Discounts spread across obligations
5. Recognise revenue	When (or as) control transfers — at a point in time or over time	Delivery, acceptance, progress of work

Point in time vs over time

Revenue is recognised **over time** if one of three criteria is met: the customer simultaneously receives and consumes the benefit (e.g., services, cleaning, SaaS); the work creates or enhances an asset the customer controls (e.g., building on the customer's land); or the asset has no alternative use to the seller and the

seller has an enforceable right to payment for work done to date (e.g., some custom manufacturing). Otherwise revenue is recognised at a **point in time** when control passes, considering indicators: right to payment, legal title, physical possession, risks and rewards, customer acceptance.

WORKED EXAMPLE · Bundled contract (fictional)

A supplier sells a cold-storage system for $500,000 including installation and 2 years of maintenance. Standalone selling prices: equipment $440,000; installation $40,000; maintenance $80,000 (total $560,000). The $60,000 discount is allocated proportionally.

Obligation	SSP	Allocation %	Allocated price	Recognised
Equipment	440,000	78.57%	392,857	On delivery (point in time)
Installation	40,000	7.14%	35,714	When installed (or over the short installation period)
Maintenance	80,000	14.29%	71,429	Evenly over 24 months: ≈ 2,976 per month
Total	560,000	100%	500,000	

If the customer pays $500,000 upfront, the unearned maintenance portion sits in contract liabilities (deferred revenue).

Incoterms, delivery and revenue timing

For goods sales, the point at which control transfers is often linked to the delivery terms, although Incoterms deal with risk and costs, not ownership, and the full contract must be considered.

Delivery term	Typical indication of control transfer	Revenue timing (typical)
EXW	Goods placed at buyer's disposal at seller's premises	When made available and collected/at buyer's disposal
FOB / FCA	Delivery at origin (on board / to carrier)	At shipment
CIF / CFR / CIP / CPT	Risk passes at origin, seller arranges carriage	Often at shipment; freight/insurance may be a separate service obligation recognised over the voyage under IFRS 15 analysis
DAP / DPU / DDP	Delivery at destination	On arrival/delivery at destination

COMMON MISTAKES · Revenue-recognition red flags

- Large sales booked in the last days of a quarter ("channel stuffing"), followed by returns.
- Bill-and-hold arrangements without substantive reasons.
- Revenue recognised for goods shipped on consignment (the distributor has not bought them).
- Gross revenue reported where the company is only an agent.
- Percentage-of-completion estimates that keep rising while cash lags.

20.3 Principal vs agent

If a company controls the goods or services before transfer to the customer, it is a **principal** and reports revenue **gross**; if it arranges for another party to provide them, it is an **agent** and reports only its commission or margin **net**. Indicators of principal status: primary responsibility for fulfilment, inventory risk, discretion in setting price.

WORKED EXAMPLE · Marketplace vs reseller

A platform sells a $1,000 holiday package supplied by a hotel, keeping $150. If the platform only arranges the booking (agent), revenue = **$150**. If it buys room allotments in advance at its own risk and resells them (principal), revenue = **$1,000** and cost of sales = $850. The profit is identical; revenue differs by a factor of almost 7 — another reason to compare companies on gross profit or margins rather than headline revenue.

20.4 Expense recognition and matching

Expenses are recognised when the related economic benefits are consumed: cost of goods sold when the revenue is recognised; depreciation over useful life; salaries in the period worked; marketing generally when incurred (even if benefits are long-term — advertising is not capitalised under IFRS); research costs expensed; development costs capitalised under IAS 38 only when strict criteria are met (US GAAP generally expenses R&D, with specific exceptions such as certain software costs).

Accruals, prepayments and cut-off

Situation	Adjustment	Entry
Expense incurred, invoice not yet received	Accrued expense	Dr Expense / Cr Accrued expenses
Expense paid in advance	Prepayment	Dr Prepayments / Cr Cash; then release monthly
Revenue earned, not yet invoiced	Accrued income / contract asset	Dr Accrued income / Cr Revenue
Cash received before earning	Deferred revenue / contract liability	Dr Cash / Cr Contract liability

Cut-off testing verifies that transactions are recorded in the correct period — auditors examine deliveries and invoices just before and after the year-end.

PRACTICE · Practice questions

1. A company sells a machine for $200,000 on 20 December under DAP terms; it arrives at the customer on 8 January. Year-end is 31 December. When is revenue recognised?
2. A software company sells a 3-year subscription for $36,000 paid upfront on 1 July. Year-end 31 December. Compute revenue for the year and deferred revenue at year-end, splitting current and non-current.
3. A travel agency sells $2m of flights as agent, earning 6% commission. What revenue does it report?
4. A distributor gives a customer a 3% rebate if annual purchases exceed $1m. By September purchases are $900k and the customer is very likely to exceed $1m. How should revenue be measured?

ANSWER · Answers to the practice questions

1. Under DAP the seller delivers at destination; control typically transfers on arrival → revenue in **January** (next year), not December. In December the machine remains in the seller's inventory (in transit).
2. Monthly revenue = 1,000. Revenue this year (July–December) = **$6,000**. Deferred revenue at 31 Dec = **$30,000**: current (next 12 months) $12,000; non-current $18,000.
3. Net commission: 6% × 2m = **$120,000**.
4. Variable consideration: since the rebate is highly likely, revenue on each sale should be recognised net of the expected 3% rebate (with a refund liability/accrued rebate), not gross with a lump-sum rebate later.

CHAPTER 21

Assets and Liabilities in Detail

LEARNING OBJECTIVES · In this chapter

The accounting for the balance-sheet items business people deal with most: inventory, receivables and payables, property and equipment, intangible assets and goodwill, impairment, provisions and contingencies, leases, and deferred tax — each with the commercial consequences.

21.1 Inventory (IAS 2)

Cost of inventory includes purchase price, import duties and non-recoverable taxes, transport and handling, and other costs directly attributable to bringing goods to their present location and condition; trade discounts and rebates are deducted. For manufacturers, cost includes conversion costs (direct labour and a systematic allocation of production overheads based on normal capacity). Excluded: abnormal waste, storage (unless necessary in production), administrative overheads, selling costs.

Cost formulas: IFRS permits FIFO and weighted average (and specific identification for non-interchangeable items). LIFO is prohibited under IFRS but permitted under US GAAP.

Measurement: lower of cost and **net realisable value (NRV)** = estimated selling price − costs to complete − costs to sell. Write-downs to NRV are expensed; IFRS allows reversal if NRV recovers (US GAAP does not reverse write-downs for most inventory).

WORKED EXAMPLE · FIFO vs weighted average (fictional)

Date	Event	Units	Unit cost
1 Mar	Opening stock	1,000	$10.00
10 Mar	Purchase	2,000	$11.50
20 Mar	Sale	2,200	—

FIFO: COGS = 1,000 × 10.00 + 1,200 × 11.50 = **$23,800**; closing stock 800 × 11.50 = **$9,200**.

Weighted average: average cost = (10,000 + 23,000) ÷ 3,000 = $11.00; COGS = 2,200 × 11.00 = **$24,200**; closing stock 800 × 11.00 = **$8,800**.

With rising prices, FIFO gives lower COGS and higher profit (and closing stock closer to current cost). The cash flows are identical; only the allocation between periods changes (and possibly tax).

WORKED EXAMPLE · NRV write-down

800 units carried at $11.50 ($9,200). A competitor's launch drops the selling price to $12.00; selling costs are $1.20 per unit → NRV = $10.80. Write-down = 800 × 0.70 = **$560**: Dr Cost of sales (inventory write-down) 560 / Cr Inventory 560.

Commercial point: slow-moving and obsolete stock is one of the most common sources of overstated assets in trading companies. Lenders and buyers ask for an **inventory ageing** and the policy for provisions on old stock (e.g., 50% for items older than 12 months, 100% beyond 24).

21.2 Trade receivables and expected credit losses

Receivables are measured at amortised cost less a **loss allowance**. Under IFRS 9, trade receivables without a significant financing component use the **simplified approach**: lifetime expected credit losses (ECL), often estimated with a **provision matrix** based on historical loss rates adjusted for forward-looking information.

WORKED EXAMPLE · Provision matrix (fictional distributor)

Ageing bucket	Balance ($)	Expected loss rate	Allowance ($)
Not yet due	1,400,000	0.5%	7,000
1–30 days overdue	520,000	2%	10,400
31–60 days	260,000	6%	15,600
61–90 days	140,000	15%	21,000
Over 90 days	180,000	45%	81,000
Total	**2,500,000**		**135,000**

If the opening allowance was $95,000, the period charge is $40,000: Dr Impairment loss on receivables 40,000 / Cr Loss allowance 40,000. When a specific debt is confirmed uncollectable ($22,000): Dr Loss allowance 22,000 / Cr Trade receivables 22,000 (no further P&L effect).

21.3 Trade payables, accruals and supplier finance

Payables are generally measured at the invoice amount. Watch for:

- **Accruals** for goods received not invoiced (GRNI) and services consumed.
- **Supplier finance arrangements** (reverse factoring): a bank pays suppliers early and the buyer pays the bank later. IFRS and US GAAP now require specific disclosures because such arrangements can make debt-like funding look like ordinary payables; analysts may reclassify them as debt.
- **Customer advances** are contract liabilities, not payables.

21.4 Property, plant and equipment (IAS 16)

Recognition: an item is capitalised if it is probable that future economic benefits will flow and cost can be measured reliably. Cost includes purchase price, duties, directly attributable costs (delivery, installation, professional fees, testing) and estimated dismantling/restoration costs. Day-to-day servicing is expensed.

Depreciation methods:

FORMULA · Depreciation

Straight-line: annual depreciation = (Cost − Residual value) ÷ Useful life

Declining balance: depreciation = Rate × Opening net book value (rate e.g. 2 ÷ useful life for double-declining)

Units of production: depreciation = (Cost − Residual) × Units this period ÷ Total expected units

WORKED EXAMPLE · Three methods for a $100,000 truck (residual $10,000, 5 years or 500,000 km)

Year	Straight-line	Double-declining (40%)	Units (km driven)
1	18,000	40,000	120,000 km → 21,600
2	18,000	24,000	110,000 km → 19,800

Year	Straight-line	Double-declining (40%)	Units (km driven)
3	18,000	14,400	100,000 km → 18,000
4	18,000	8,640	90,000 km → 16,200
5	18,000	2,960 (to reach residual)	80,000 km → 14,400
Total	90,000	90,000	90,000

Same total; different timing. Management's choice of useful life and method changes reported profit — another place analysts look for aggressive accounting (e.g., suddenly extending useful lives).

Component accounting: significant parts with different lives are depreciated separately (aircraft engines vs airframe; ship dry-docking costs).

Disposal: gain or loss = proceeds − net book value. Example: truck with NBV $46,000 sold for $50,000 → gain $4,000 (Dr Cash 50,000; Dr Accumulated depreciation 54,000; Cr Truck at cost 100,000; Cr Gain on disposal 4,000).

Revaluation model: IFRS permits revaluing classes of PP&E to fair value (common for land and buildings in some regions), with gains to a revaluation reserve in equity. US GAAP does not permit upward revaluation. Lenders often exclude revaluation reserves when computing tangible net worth for covenants.

21.5 Intangible assets and goodwill (IAS 38, IFRS 3)

Intangibles: software, licences, patents, trademarks, customer relationships (when acquired), development costs (when IAS 38 criteria are met: technical feasibility, intention and ability to complete and use or sell, probable future benefits, resources available, reliable measurement). Research is always expensed. Internally generated brands, customer lists and similar items are not recognised.

Finite-life intangibles are amortised; indefinite-life intangibles and **goodwill** are not amortised under IFRS but tested for impairment at least annually.

21.6 Impairment (IAS 36)

An asset (or cash-generating unit, CGU) is impaired when its carrying amount exceeds its **recoverable amount** = higher of fair value less costs of disposal and value in use (present value of future cash flows). Impairment losses are expensed; goodwill impairments can never be reversed; other impairments may be reversed under IFRS (not under US GAAP for assets held and used).

Commercial meaning: large impairments tell you past investment decisions have failed. They are non-cash in the year recognised, but the cash was lost earlier when the asset was bought.

21.7 Provisions, contingent liabilities and contingent assets (IAS 37)

DEFINITION · Recognition test for a provision

Recognise a provision when (1) there is a present obligation (legal or constructive) from a past event, (2) an outflow of resources is probable (more likely than not), and (3) a reliable estimate can be made. Otherwise, disclose a **contingent liability** unless the possibility of outflow is remote. **Contingent assets** (e.g., a probable insurance recovery) are not recognised until virtually certain; they are disclosed when an inflow is probable.

Situation	Treatment
Warranties on products sold (historical claims 1.5% of sales)	Provision = expected cost of claims

Situation	Treatment
Lawsuit: lawyers estimate 70% probability of losing $400k	Provision (best estimate, ~$400k)
Lawsuit: 30% probability of losing	Contingent liability — disclose
Lawsuit: remote probability	No disclosure required
Restructuring announced with a detailed formal plan	Provision for direct costs
Future operating losses	No provision
Bank guarantee issued for a subsidiary	Contingent liability (or financial guarantee accounting)
Onerous contract (unavoidable costs exceed benefits)	Provision for the unavoidable loss
End-of-service benefits for employees (common in GCC labour laws)	Employee benefit obligation (IAS 19), not IAS 37

21.8 Leases (IFRS 16)

For lessees, IFRS 16 (effective 2019) brings almost all leases onto the balance sheet: a **right-of-use (ROU) asset** and a **lease liability** equal to the present value of lease payments. Exemptions: short-term leases (≤ 12 months) and low-value assets. The lessee then recognises depreciation of the ROU asset and interest on the liability instead of a rent expense. US GAAP (ASC 842) also puts leases on the balance sheet but keeps a straight-line single lease cost for operating leases in the income statement.

WORKED EXAMPLE · Warehouse lease under IFRS 16

5-year lease, $100,000 paid at the end of each year, incremental borrowing rate 7%.

- Lease liability at start = 100,000 × annuity factor (5 years, 7%) = 100,000 × 4.1002 = **$410,020**; ROU asset the same.
- Year 1: interest = 7% × 410,020 = **$28,701**; depreciation = 410,020 ÷ 5 = **$82,004**; total expense $110,705 (vs $100,000 rent previously); liability after payment = 410,020 + 28,701 − 100,000 = $338,721.
- **EBITDA rises by $100,000** (rent removed; depreciation and interest are below EBITDA), net debt rises by $410,020, and operating cash flow rises because principal repayments are financing cash flows.

That is why bank covenants and analysts specify whether figures are “pre-IFRS 16” or “post-IFRS 16”.

21.9 Deferred tax (IAS 12) — the essentials

Accounting profit and taxable profit differ. **Temporary differences** (e.g., tax depreciation faster than accounting depreciation) create deferred tax liabilities or assets that reverse later. Deferred tax assets from tax losses are recognised only if future taxable profits are probable. Business implication: a company may show a tax expense but pay little cash tax (or vice versa); analysts look at **cash tax paid** in the cash flow statement.

PRACTICE · Practice questions

1. A company bought 500 units at $20 and later 500 at $26, then sold 600. Compute COGS under FIFO and weighted average.
2. A $60,000 machine, residual $6,000, life 6 years, is sold at the end of year 3 for $28,000 (straight-line). Compute the gain or loss.
3. Classify: (a) a probable $2m environmental clean-up obligation for contamination caused last year; (b) a possible patent infringement claim; (c) expected losses next year from a weak product line.

4. A retailer signs a 10-year store lease at $200,000 per year (paid in arrears) with an incremental borrowing rate of 8%. Compute the initial lease liability (annuity factor 6.7101) and the year-1 interest and depreciation. How does EBITDA change versus the old treatment?

ANSWER · Answers to the practice questions

1. FIFO: 500 × 20 + 100 × 26 = **$12,600**. Weighted average: (10,000 + 13,000) ÷ 1,000 = 23 → 600 × 23 = **$13,800**.
2. Annual depreciation = (60,000 − 6,000) ÷ 6 = 9,000; NBV after 3 years = 60,000 − 27,000 = 33,000; loss = 28,000 − 33,000 = **−$5,000**.
3. (a) Provision (present obligation, probable, estimable). (b) Contingent liability — disclose if not remote. (c) No provision — future operating losses are not provided for (unless an onerous contract exists).
4. Liability = 200,000 × 6.7101 = **$1,342,020**. Year-1 interest = 8% × 1,342,020 = **$107,362**; depreciation = 1,342,020 ÷ 10 = **$134,202**. EBITDA **increases by $200,000** because rent is no longer an operating expense.

CHAPTER 22

The Three Financial Statements and a Complete Worked Example

LEARNING OBJECTIVES · In this chapter

The income statement, balance sheet (statement of financial position) and cash flow statement in depth; how they link; how single transactions move all three; and a complete worked example from opening balances through journal entries to final statements.

22.1 The income statement (statement of profit or loss)

Reports performance over a period. IFRS allows presentation of expenses **by function** (cost of sales, distribution, administrative) or **by nature** (materials, staff costs, depreciation). IFRS 18 *Presentation and Disclosure in Financial Statements*, issued in 2024 and effective for annual periods beginning on or after 1 January 2027, will replace IAS 1 and introduce defined subtotals (operating profit, profit before financing and income taxes) and disclosure of management-defined performance measures; verify adoption status for any company you analyse.

Line	What to look for
Revenue	Growth, segment and geographic split, recognition policy
Cost of sales / gross profit	Gross margin trend; input-cost pressures
Distribution and administrative expenses	Operating leverage; one-offs
Other operating income/expense	Gains on disposals, FX, impairments — sustainable?
Operating profit	Core profitability
Finance income/costs	Debt burden, interest rate exposure, FX on borrowings
Share of profit of associates	Non-controlled investments
Income tax	Effective tax rate vs statutory
Profit for the year; attributable to owners vs NCI	What belongs to the parent's shareholders
Other comprehensive income	Revaluations, FX translation, hedge reserves — not in profit

22.2 The balance sheet (statement of financial position)

A snapshot at a date: what the entity controls (assets), what it owes (liabilities), and the residual (equity), split into current and non-current.

Section	Typical lines	Questions to ask
Non-current assets	PP&E, ROU assets, intangibles, goodwill, investments, deferred tax assets	Age of assets? Goodwill size relative to equity?
Current assets	Inventory, trade receivables, prepayments, contract assets, cash	Quality and ageing; restricted cash?
Current liabilities	Trade payables, accruals, short-term borrowings, current portion of long-term debt, lease liabilities, contract liabilities, tax	Refinancing needs within 12 months?
Non-current liabilities	Long-term borrowings, lease liabilities, provisions, employee benefits, deferred tax	Maturity profile; covenants

Section	Typical lines	Questions to ask
Equity	Share capital, reserves, retained earnings, NCI	Tangible net worth after goodwill

22.3 The cash flow statement

Reports cash movements in three categories (IAS 7):

- **Operating activities:** cash from trading. Usually presented with the **indirect method**: start from profit, add back non-cash items, adjust for working-capital changes. (The direct method shows receipts and payments; encouraged by IAS 7 but less common.)
- **Investing activities:** capex, acquisitions, disposals, investments.
- **Financing activities:** borrowings and repayments, share issues, dividends, lease principal.

Interest and dividends paid/received may be classified in operating or financing/investing under current IAS 7 (choice applied consistently); US GAAP generally puts interest paid in operating. IFRS 18 introduces consequential amendments that reduce these choices for most companies from 2027.

22.4 How the statements link

1. **Net profit** from the income statement flows into **retained earnings** (balance sheet) and is the starting point of the indirect cash flow statement.
2. **Changes in balance-sheet items** (receivables, inventory, payables, fixed assets, debt, equity) explain the cash flows.
3. **Ending cash** in the cash flow statement equals cash on the balance sheet.
4. **Depreciation** reduces PP&E (balance sheet), is an expense (income statement) and is added back (cash flow).

22.5 One transaction, three statements

Assume a 25% tax rate where relevant and that tax is paid in cash at once (to keep the logic visible).

Transaction	Income statement	Balance sheet	Cash flow statement
Sell goods on credit for $100 (cost $60)	Revenue +100, COGS −60, tax −10 → NI +30	Receivables +100, inventory −60, cash −10 (tax), RE +30	CFO: NI +30, Δreceivables −100, Δinventory +60 → −10
Customer pays the $100	No effect	Cash +100, receivables −100	CFO +100 (receivables decrease)
Buy equipment for $1,000 cash	No immediate effect	PP&E +1,000, cash −1,000	CFI −1,000
Depreciation of $200	Expense −200, tax saving +50 → NI −150	PP&E −200, cash +50, RE −150	CFO: NI −150 + D&A 200 = +50
Borrow $500	No effect	Cash +500, debt +500	CFF +500
Pay interest of $40	Expense −40, tax +10 → NI −30	Cash −30, RE −30	CFO −30 (if interest in operating)
Repay $100 of loan principal	No effect	Cash −100, debt −100	CFF −100
Inventory write-down of $30	Expense −30, tax +7.5 → NI −22.5	Inventory −30, cash +7.5, RE −22.5	CFO: NI −22.5 + non-cash 30 = +7.5
Pay dividend of $50	No effect (not an expense)	Cash −50, RE −50	CFF −50

Transaction	Income statement	Balance sheet	Cash flow statement
Issue shares for $300	No effect	Cash +300, share capital +300	CFF +300
Prepay a year's insurance $120	No effect	Prepayments +120, cash −120	CFO −120 (Δprepayments)

PROFESSIONAL INSIGHT · The classic interview question

"Depreciation increases by $10. Walk me through the three statements" (25% tax). Income statement: pre-tax profit −10, tax −2.5, net income −7.5. Cash flow: net income −7.5, add back depreciation +10 → cash +2.5 (lower tax paid). Balance sheet: PP&E −10, cash +2.5 → assets −7.5; retained earnings −7.5 → balanced. A finance graduate who can do this for any item in seconds has internalised accounting.

22.6 Complete worked example: Levant Office Solutions Ltd (fictional)

Levant Office Solutions distributes office furniture and equipment. All amounts in USD. Financial year 1 January – 31 December.

Opening balance sheet at 1 January

Assets	$	Liabilities and equity	$
Cash	50,000	Trade payables	70,000
Trade receivables	80,000	Accrued expenses	10,000
Inventory	120,000	Bank loan	150,000
PP&E at cost	300,000	Share capital	100,000
Accumulated depreciation	(60,000)	Retained earnings	160,000
Total	**490,000**	**Total**	**490,000**

Transactions during the year (summarised)

1. Credit sales of $1,200,000.
2. Cash sales of $100,000.
3. Inventory purchased on credit: $820,000.
4. Closing inventory counted at $150,000 (so COGS = 120,000 + 820,000 − 150,000 = $790,000).
5. Cash collected from credit customers: $1,170,000.
6. Cash paid to suppliers: $800,000.
7. Salaries paid: $180,000.
8. Rent paid: $48,000, of which $4,000 relates to next January.
9. The opening accrued expenses of $10,000 were paid. Other operating expenses for the year were $60,000, of which $52,000 was paid and $8,000 is unpaid at year-end.
10. New equipment purchased for cash on 1 July: $80,000.
11. Depreciation: 10% per year straight-line on cost; half a year on assets bought mid-year.
12. Loan principal repaid $30,000; interest paid $12,000.
13. Income tax at 15% of profit before tax, payable next year.
14. Dividend paid: $20,000.

Journal entries (summarised)

#	Debit	Credit	Amount
1	Trade receivables	Revenue	1,200,000
2	Cash	Revenue	100,000
3	Inventory	Trade payables	820,000
4	Cost of goods sold	Inventory	790,000
5	Cash	Trade receivables	1,170,000
6	Trade payables	Cash	800,000
7	Salaries expense	Cash	180,000
8a	Rent expense	Cash	44,000
8b	Prepayments	Cash	4,000
9a	Accrued expenses	Cash	10,000
9b	Other operating expenses	Cash	52,000
9c	Other operating expenses	Accrued expenses	8,000
10	PP&E at cost	Cash	80,000
11	Depreciation expense	Accumulated depreciation	34,000
12a	Bank loan	Cash	30,000
12b	Interest expense	Cash	12,000
13	Income tax expense	Tax payable	27,000
14	Retained earnings (dividends)	Cash	20,000

Depreciation: 10% × 300,000 = 30,000 on existing assets; 10% × 80,000 × ½ = 4,000 on new equipment; total **34,000**. Tax: computed below.

Income statement for the year

Line	$
Revenue (1,200,000 + 100,000)	1,300,000
Cost of goods sold	(790,000)
Gross profit (39.2%)	**510,000**
Salaries	(180,000)
Rent	(44,000)
Other operating expenses	(60,000)
EBITDA (17.4%)	**226,000**
Depreciation	(34,000)
Operating profit / EBIT (14.8%)	**192,000**
Interest expense	(12,000)
Profit before tax	**180,000**
Income tax (15%)	(27,000)
Net income (11.8%)	**153,000**

Closing balance sheet at 31 December

Assets	$	Liabilities and equity	$
Cash	88,000	Trade payables (70 + 820 − 800)	90,000
Trade receivables (80 + 1,200 − 1,170)	110,000	Accrued expenses	8,000
Inventory	150,000	Tax payable	27,000
Prepayments	4,000	Bank loan (150 − 30)	120,000
PP&E at cost (300 + 80)	380,000	Share capital	100,000
Accumulated depreciation (60 + 34)	(94,000)	Retained earnings (160 + 153 − 20)	293,000
Total	**638,000**	**Total**	**638,000**

Cash check: 50 + 100 + 1,170 − 800 − 180 − 48 − 10 − 52 − 80 − 30 − 12 − 20 = **88** thousand.

Cash flow statement — indirect method

Line	$
Net income	153,000
Add: depreciation	34,000
Add: increase in tax payable (tax expensed, not yet paid)	27,000
Increase in trade receivables	(30,000)
Increase in inventory	(30,000)
Increase in prepayments	(4,000)
Increase in trade payables	20,000
Decrease in accrued expenses	(2,000)
Net cash from operating activities	**168,000**
Purchase of equipment	(80,000)
Net cash used in investing activities	**(80,000)**
Repayment of bank loan	(30,000)
Dividends paid	(20,000)
Net cash used in financing activities	**(50,000)**
Net increase in cash	**38,000**
Cash at 1 January	50,000
Cash at 31 December	**88,000**

(Interest paid of $12,000 is included in operating activities here, inside net income.)

Cash flow statement — direct method (operating section)

Line	$
Cash received from customers (100,000 + 1,170,000)	1,270,000
Cash paid to suppliers	(800,000)
Cash paid to employees	(180,000)
Rent paid	(48,000)
Other operating expenses paid (52,000 + 10,000)	(62,000)
Interest paid	(12,000)

Line	$
Net cash from operating activities	**168,000**

Interpretation

- **Profit to cash conversion:** operating cash flow 168k vs net income 153k — healthy, helped by unpaid tax (27k) that will be paid next year. Excluding that timing benefit, conversion is 141k ÷ 153k ≈ 92%.
- **Working capital** absorbed 46k (receivables +30k, inventory +30k, prepayments +4k, payables −20k, accruals +2k), consistent with growth.
- **Free cash flow** (CFO − capex) = 88k; it covered dividends (20k) and debt repayment (30k), and cash rose 38k.
- **Ratios:** current ratio = (88 + 110 + 150 + 4) ÷ (90 + 8 + 27 + current portion of loan, say 30) = 352 ÷ 155 = 2.3×; net debt = 120 − 88 = 32k; net debt/EBITDA ≈ 0.14×; ROE = 153 ÷ average equity ((260 + 393) ÷ 2 = 326.5) ≈ 47%.
- **Questions an analyst would ask:** Is the 39% gross margin sustainable? Is the inventory count reliable? What are receivables' ageing and concentration? Was the tax computation correct (any non-deductible expenses)?

بالعربية — ملخص المثال التطبيقي

بدأت الشركة السنة بنقد 50 ألف دولار، وحققت صافي ربح 153 ألفاً، وتدفقاً نقدياً تشغيلياً 168 ألفاً. اشترت معدات بـ80 ألفاً، وسددت 30 ألفاً من القرض، ووزعت أرباحاً بقيمة 20 ألفاً، فارتفع النقد إلى 88 ألفاً. الربط بين القوائم الثلاث واضح: صافي الربح ينتقل إلى الأرباح المحتجزة في الميزانية، ونقطة بداية قائمة التدفقات النقدية هي صافي الربح، ورصيد النقد في نهاية قائمة التدفقات يساوي رصيد النقد في الميزانية.

22.7 Accounting profit vs cash generated: a summary

Reason profit ≠ cash	Direction	Example
Non-cash expenses (depreciation, amortisation, impairments, provisions)	Cash > profit	Depreciation 34k
Working-capital investment	Cash < profit when receivables/inventory rise	+30k receivables
Capex	Cash < profit (capex not expensed)	80k equipment
Timing of tax payments	Either	Tax payable 27k
Debt principal flows	Do not affect profit	30k repayment
Dividends	Do not affect profit	20k
Non-cash revenue (e.g., fair-value gains)	Profit > cash	Revaluation gains
Customer prepayments	Cash > profit	Subscriptions

PRACTICE · Practice questions

1. Using the Levant Office Solutions example, suppose credit customers had paid only $1,100,000 (instead of $1,170,000). What changes in each statement?
2. Walk through the three statements for: a $50,000 inventory purchase on credit; then the sale of half of it for $40,000 cash (tax 15%, paid immediately).
3. A company reports net income of $5m and CFO of $1m. List the balance-sheet lines you would examine first.

ANSWER · Answers to the practice questions

1. Income statement: no change (revenue already recognised). Balance sheet: receivables $180,000 (+70k), cash $18,000 (−70k). Cash flow: increase in receivables (100,000) instead of (30,000); CFO $98,000; net increase in cash −$32,000 (from 50k to 18k). The company would need to use its overdraft — profit unchanged, cash much weaker.
2. Purchase: inventory +50k, payables +50k; no IS or cash effect (CFO: Δinventory −50k, Δpayables +50k = 0). Sale: revenue +40k, COGS −25k, pre-tax +15k, tax −2.25k → NI +12.75k; cash +40k − 2.25k = +37.75k; inventory −25k; RE +12.75k. Cash flow: NI 12.75 + Δinventory +25 = 37.75 ✓.
3. Trade receivables and contract assets (rising? overdue?), inventory (build-up, obsolescence), payables (lower — paying suppliers faster?), prepayments and advances to suppliers, capitalised costs and other receivables, and non-cash gains in profit (fair-value gains, disposals).

CHAPTER 23

Consolidation, IFRS vs US GAAP and the Limits of Financial Reporting

LEARNING OBJECTIVES · In this chapter

How group accounts are prepared; the main differences between IFRS and US GAAP that a business professional should recognise; and what financial statements cannot tell you.

23.1 Consolidated financial statements (IFRS 10)

A parent that **controls** subsidiaries presents consolidated statements as if the group were a single economic entity. Control exists when the investor has power over the investee, exposure to variable returns, and the ability to use its power to affect those returns — usually, but not always, through more than 50% of voting rights.

Relationship	Typical ownership	Accounting
Subsidiary (control)	> 50% or control by agreement	Full consolidation; non-controlling interest (NCI) shown in equity
Joint venture (joint control, rights to net assets)	Often 50/50	Equity method (IFRS 11)
Joint operation (rights to assets and obligations for liabilities)	Varies	Recognise share of assets, liabilities, revenue, expenses
Associate (significant influence)	Typically 20–50%	Equity method (IAS 28): one line in balance sheet and income statement
Financial investment	< 20%, no influence	Fair value (IFRS 9)

Consolidation steps

1. Add together line by line the parent's and subsidiaries' assets, liabilities, income and expenses (after aligning accounting policies and translating foreign subsidiaries into the presentation currency).
2. Eliminate the parent's investment against the subsidiary's equity at acquisition; recognise goodwill and NCI.
3. Eliminate **intra-group** balances (receivables/payables, loans) and transactions (sales, dividends, management fees), and unrealised profits in inventory sold within the group.
4. Allocate profit and equity between owners of the parent and NCI.

WORKED EXAMPLE · Simple consolidation with NCI (fictional)

Parent P acquires 80% of S for $800k when S's net assets (fair value) are $750k. NCI measured at its share of net assets: 20% × 750 = $150k. Goodwill = 800 + 150 − 750 = **$200k**.

During the year S earns $100k profit and sells goods to P for $60k with a $15k margin; P still holds half of them at year-end. Intra-group sale of $60k is eliminated from revenue and COGS; unrealised profit of $7.5k is eliminated from inventory. S's adjusted profit 92.5k → NCI share 18.5k; parent share 74k.

COMMON MISTAKES · Why consolidation matters to bankers and investors

- **Structural subordination:** a lender to the holding company is paid only from dividends upstreamed by subsidiaries, after subsidiaries' own creditors.

- **Trapped cash:** consolidated cash may sit in countries with exchange controls.
- **Intra-group transactions** can inflate the revenue of individual entities; always ask for consolidated and standalone figures.
- **NCI:** consolidated EBITDA includes 100% of subsidiaries' EBITDA even if the parent owns 51%.

23.2 Foreign currency (IAS 21) — essentials

Each entity has a **functional currency** (the currency of its primary economic environment). Foreign-currency transactions are recorded at the spot rate; monetary items (receivables, payables, loans) are retranslated at closing rates, with differences in profit or loss. Foreign subsidiaries are translated into the group's presentation currency (assets/liabilities at closing rate, income at average rates), with differences in other comprehensive income (the translation reserve). Hyperinflationary economies require IAS 29 restatement — relevant in several countries in recent years.

23.3 IFRS vs US GAAP: differences a professional should recognise

Topic	IFRS	US GAAP
Framework style	More principles-based	More rules and industry guidance
Inventory cost	FIFO or weighted average; LIFO prohibited	LIFO permitted
Inventory write-down reversals	Required when NRV recovers	Not permitted
Development costs	Capitalised when criteria met	Generally expensed (exceptions, e.g., certain software)
PP&E revaluation	Permitted (revaluation model)	Not permitted
Impairment reversals (other than goodwill)	Permitted	Not permitted for assets held and used
Leases (lessee)	Single model: depreciation + interest	Finance vs operating leases; operating leases show straight-line lease cost
Interest paid in cash flow	Operating or financing (until IFRS 18 changes)	Operating
Extraordinary items	Prohibited	Eliminated
Debt covenant breach before year-end	Classification depends on rights at reporting date (IAS 1 amendments)	Specific rules on waivers
Financial statement titles	Statement of financial position; statement of profit or loss	Balance sheet; income statement

Revenue recognition (IFRS 15 / ASC 606) is largely converged; differences remain in details.

23.4 The limits of financial reporting

Financial statements are indispensable but incomplete:

1. **Historical:** they report the past, often 2–4 months after period-end.
2. **Estimates and judgement:** useful lives, provisions, expected credit losses, fair values, revenue over time.
3. **Unrecognised value:** internally generated brands, customer relationships, talent, data and culture do not appear.
4. **Management incentives:** choices can smooth or flatter results within the rules (and fraud can occur outside them).

5. **Aggregation:** consolidated numbers hide segment problems.
6. **Off-balance-sheet exposures:** guarantees, commitments, some supplier finance, contingent liabilities appear only in notes.
7. **Non-financial risks:** climate, cyber, regulatory, key-person risk — partly addressed by sustainability reporting (e.g., the ISSB's IFRS S1 and S2 standards, issued 2023 and adopted by an increasing number of jurisdictions).
8. **Comparability limits:** different standards, policies, and non-GAAP "adjusted" metrics.

PROFESSIONAL INSIGHT · How professionals compensate

They read the **notes** (accounting policies, segment information, related parties, contingencies, debt maturities, subsequent events), the **auditor's report** (including key audit matters), management commentary, and they triangulate with non-financial data: customer interviews, site visits, industry data, payment behaviour, credit bureau reports.

PRACTICE · Practice questions

1. A group owns 60% of a subsidiary with EBITDA of $10m and net debt of $30m; the parent standalone has EBITDA $5m and net debt $20m. Compute consolidated EBITDA and net debt and the parent's "look-through" proportional figures. Which matters for a lender to the parent?
2. Name four items you would look for in the notes before lending to a company.
3. A company switches from expensing to capitalising development costs. What happens to EBITDA, net income and operating cash flow?

ANSWER · Answers to the practice questions

1. Consolidated EBITDA = **$15m**; net debt **$50m** (leverage 3.3×). Proportional: EBITDA 5 + 6 = $11m; net debt 20 + 18 = $38m (3.5×). A lender to the parent cares about the parent's own cash and its access to subsidiary cash (dividends 60% after the subsidiary's debt service), and about structural subordination — consolidated ratios can overstate the parent lender's position.
2. Debt maturities and covenants; contingent liabilities and guarantees; related-party transactions; segment results; accounting policy changes; subsequent events; lease commitments; pledged assets.
3. EBITDA rises (costs move from operating expenses to an asset, later amortised below EBITDA); net income rises initially; operating cash flow rises while investing cash flow falls by the same amount — total cash is unchanged.

CHAPTER 24

Management Accounting, Cost Accounting, Budgeting and Variance Analysis

LEARNING OBJECTIVES · In this chapter

Accounting for decision-making: cost behaviour and costing methods, contribution analysis, relevant costs, budgeting and forecasting, and variance analysis that explains why results differ from plan.

24.1 Financial vs management accounting

	Financial accounting	Management accounting
Users	External: investors, lenders, regulators, tax	Internal: managers, board
Rules	IFRS/GAAP, law	Whatever is useful
Focus	Entity as a whole, historical	Products, customers, projects, future
Frequency	Quarterly/annual	Daily to monthly
Precision	High, audited	Timely and relevant beats precise

24.2 Costing methods

- **Job costing:** costs accumulated per order or project (construction, consulting, custom manufacturing).
- **Process costing:** costs averaged over continuous output (cement, beverages, chemicals).
- **Absorption costing:** product cost includes fixed production overheads (required for external inventory valuation).
- **Marginal (variable) costing:** only variable costs are product costs; fixed costs are period costs — better for short-term decisions.
- **Activity-based costing (ABC):** overheads assigned according to the activities products/customers consume (orders, deliveries, set-ups). ABC often reveals that small, complex customers are unprofitable.
- **Standard costing:** predetermined costs per unit used for planning and variance analysis.

WORKED EXAMPLE · Customer profitability with ABC (fictional distributor)

	Customer A (hotel chain)	Customer B (small restaurants, 40 accounts)
Revenue	$1,000,000	$1,000,000
Gross margin	18% → $180,000	30% → $300,000
Orders per year (cost $45 per order processed)	120 → $5,400	4,000 → $180,000
Deliveries ($60 each)	150 → $9,000	3,500 → $210,000
Credit control and collection	$6,000	$40,000
Account management	$25,000	$20,000
Customer contribution	**$134,600**	**−$150,000**

The high-margin small customers destroy value once order and delivery costs are assigned. Actions: minimum order values, delivery charges, digital ordering, or serving them through a wholesaler.

24.3 Relevant costs for decisions

A cost is **relevant** to a decision if it is a future cash flow that differs between alternatives. Ignore sunk costs and allocated fixed costs that will not change. Include opportunity costs.

> **WORKED EXAMPLE · Special order decision**
>
> A factory with spare capacity receives a one-off export order for 10,000 units at $8.00. Full absorption cost is $9.20 (variable $6.50 + fixed overhead allocation $2.70). Relevant cost = variable $6.50 + special packaging $0.30 = $6.80 → contribution **$12,000**. Accept, provided it does not undermine domestic pricing, uses genuinely spare capacity, and the buyer's credit risk is acceptable.

24.4 Budgeting

A budget is a quantified plan for a period, usually a year, broken into months. It translates strategy into targets, allocates resources, and provides a benchmark for control.

The budget sequence for a trading or manufacturing company:

1. Sales budget (volume × price by product/customer/region) — the hardest and most important.
2. Production or purchasing budget (sales + desired closing stock − opening stock).
3. Direct cost budgets (materials, labour, freight).
4. Overhead budgets by cost centre.
5. Capex budget.
6. Budgeted income statement.
7. **Cash budget** (timing of receipts and payments) — reveals funding needs.
8. Budgeted balance sheet.

Approach	Description	Pros	Cons
Incremental	Last year + x%	Simple	Perpetuates inefficiency
Zero-based (ZBB)	Every cost justified from zero	Challenges spending	Time-consuming
Rolling forecast	Continuously extends 12–18 months ahead	Adapts to change	Requires discipline and systems
Activity-based	Based on planned activity drivers	Links cost to operations	Complex
Flexible budget	Adjusts to actual volume	Fair performance comparison	Needs cost behaviour data

> **WORKED EXAMPLE · Cash budget (fictional, three months)**
>
> Sales: Jan $400k, Feb $500k, Mar $600k; 30% cash, 70% collected next month (December sales $380k). Purchases = 70% of next month's sales, paid the month after purchase (December purchases $280k). Fixed cash costs $90k/month. Opening cash $60k; overdraft limit $200k.

$000	Jan	Feb	Mar
Cash sales (30%)	120	150	180
Collections (70% of last month)	266	280	350
Payments to suppliers (prior month's purchases)	(280)	(350)	(420)
Fixed costs	(90)	(90)	(90)
Net cash flow	16	(10)	20
Opening cash	60	76	66

$000	Jan	Feb	Mar
Closing cash	76	66	86

Purchases: Jan purchases = 70% × Feb sales = 350 (paid Feb); Feb purchases = 70% × 600 = 420 (paid Mar). Growth absorbs cash: despite rising sales, cash barely grows. If April sales jump to $900k, March purchases (630) paid in April would push cash negative — the budget gives the treasurer time to arrange funding.

24.5 Variance analysis

Variance = actual − budget (or standard). **Favourable (F)** variances increase profit; **adverse (A)** reduce it. The power of variance analysis is decomposing a variance into causes that someone can act on.

FORMULA · Sales and cost variances

Sales price variance = (Actual price − Budget price) × Actual volume
Sales volume variance (contribution basis) = (Actual volume − Budget volume) × Budget contribution per unit
Material price variance = (Actual price − Standard price) × Actual quantity purchased
Material usage variance = (Actual quantity used − Standard quantity for actual output) × Standard price
Labour rate variance = (Actual rate − Standard rate) × Actual hours
Labour efficiency variance = (Actual hours − Standard hours for actual output) × Standard rate

WORKED EXAMPLE · Variance analysis (fictional month)

Budget: 10,000 units at $50; variable cost $30 (material 2 kg × $10 + labour 0.5 h × $20); fixed costs $120k. Budget profit = 10,000 × 20 − 120,000 = $80,000.

Actual: 11,000 units sold at $48; material 23,500 kg bought and used at $10.40; labour 5,200 h at $21; fixed costs $126k. Actual profit = 528,000 − 244,400 − 109,200 − 126,000 = $48,400.

Variance	Calculation	$
Sales volume	(11,000 – 10,000) × 20	+20,000 F
Sales price	(48 – 50) × 11,000	–22,000 A
Material price	(10.40 – 10) × 23,500	–9,400 A
Material usage	(23,500 – 22,000) × 10	–15,000 A
Labour rate	(21 – 20) × 5,200	–5,200 A
Labour efficiency	(5,200 – 5,500) × 20	+6,000 F
Fixed overhead spending	126,000 – 120,000	–6,000 A
Total	80,000 → 48,400	**–31,600**

Story for management: volume beat budget but was “bought” with a 4% price cut; material usage was 6.8% over standard (quality or waste issue at the plant); labour was efficient but paid overtime rates.

PRACTICE · Practice questions

1. A product sells at $40, variable cost $26, fixed overhead allocation $9. A customer offers $30 for 2,000 units using spare capacity. Accept?
2. Budget: 5,000 units at $100, contribution $35/unit. Actual: 4,600 units at $104. Compute sales price and volume variances.

3. Why might zero-based budgeting be useful after an acquisition?

ANSWER · Answers to the practice questions

1. Relevant contribution = 30 − 26 = $4 per unit → +**$8,000**; accept if capacity is truly spare, there is no cannibalisation of $40 sales, and credit risk is acceptable. The $9 overhead allocation is irrelevant.
2. Price: (104 − 100) × 4,600 = +**$18,400 F**; volume: (4,600 − 5,000) × 35 = **−$14,000 A**; net +$4,400 F.
3. It forces the combined organisation to justify every cost from zero, exposing duplicated functions and unnecessary spending that incremental budgeting would carry over — supporting synergy delivery.

CHAPTER 25

Financial Statement Analysis and Ratios

LEARNING OBJECTIVES · In this chapter

A structured method to analyse a company's statements: profitability, efficiency, liquidity, leverage and coverage, returns, cash-flow quality, and earnings quality, with a full worked analysis.

25.1 A method, not a list of ratios

1. **Understand the business** first: model, industry, strategy, risks. Ratios without context mislead.
2. **Read the statements and notes**, including the auditor's report.
3. **Normalise:** remove one-off items, align IFRS 16 treatment, adjust for acquisitions.
4. **Compute ratios** over at least 3 years and against peers.
5. **Explain the drivers** (why did margins change?).
6. **Check cash:** does profit convert to cash?
7. **Conclude** for the decision at hand: lend, invest, partner, supply on credit.

25.2 The main ratio families

Family	Ratio	Formula	Interpretation
Growth	Revenue growth	(Rev_t ÷ Rev_t−1) − 1	Split into price, volume, mix, FX, acquisitions
Profitability	Gross margin	Gross profit ÷ revenue	Pricing power, input costs
	EBITDA margin	EBITDA ÷ revenue	Operating cash earnings proxy
	Operating (EBIT) margin	EBIT ÷ revenue	After asset consumption
	Net margin	Net income ÷ revenue	After financing and tax
Efficiency	Asset turnover	Revenue ÷ average total assets	Capital intensity
	DSO	Receivables ÷ revenue × 365	Collection speed
	DIO	Inventory ÷ COGS × 365	Stock holding
	DPO	Payables ÷ COGS × 365	Supplier credit
	Cash conversion cycle	DIO + DSO − DPO	Working-capital intensity
Liquidity	Current ratio	Current assets ÷ current liabilities	Short-term coverage (crude)
	Quick ratio	(Current assets − inventory) ÷ current liabilities	Excludes slower assets
	Cash ratio	Cash ÷ current liabilities	Strictest
Leverage	Debt / equity	Total debt ÷ equity	Balance-sheet gearing
	Net debt / EBITDA	(Debt − cash) ÷ EBITDA	Years to repay from EBITDA (rough)
	Debt / capital	Debt ÷ (debt + equity)	Capital structure
Coverage	Interest cover	EBIT (or EBITDA) ÷ interest	Ability to pay interest
	DSCR	Cash available for debt service ÷ (interest + principal)	Ability to pay all debt service
	Fixed-charge cover	(EBITDA + rent) ÷ (interest + rent)	Includes lease obligations

Family	Ratio	Formula	Interpretation
Returns	ROA	Net income ÷ average total assets	Asset productivity
	ROE	Net income ÷ average equity	Owners' return
	ROCE	EBIT ÷ (total assets − current liabilities)	Return on long-term capital
	ROIC	NOPAT ÷ invested capital	Value creation vs WACC
Cash quality	Cash conversion	CFO ÷ EBITDA (or ÷ net income)	Earnings turning into cash
	FCF margin	(CFO − capex) ÷ revenue	Cash generation
	Capex / depreciation	Capex ÷ D&A	Under-investment if persistently < 1
Market	P/E, EV/EBITDA, dividend yield	See Part X	Valuation relative to earnings

FORMULA · DuPont decomposition of ROE

ROE = Net margin × Asset turnover × Equity multiplier
= (Net income ÷ Revenue) × (Revenue ÷ Assets) × (Assets ÷ Equity)

Extended: ROE = (NI ÷ PBT) × (PBT ÷ EBIT) × (EBIT ÷ Revenue) × (Revenue ÷ Assets) × (Assets ÷ Equity)
= tax burden × interest burden × operating margin × asset turnover × leverage

25.3 Worked analysis: two fictional distributors

$m	Alpha Trading	Beta Distribution
Revenue	100	100
Gross profit	22	30
EBITDA	8	12
Depreciation	1	4
EBIT	7	8
Interest	1.5	3.5
Net income (tax 15%)	4.68	3.83
Total assets	40	70
Equity	20	18
Debt	14	42
Cash	2	4
Receivables	16	20
Inventory	12	25
Payables	8	9
CFO	6.5	4.0
Capex	1.0	6.0

Ratio	Alpha	Beta
Gross margin	22%	30%
EBITDA margin	8%	12%
Net margin	4.7%	3.8%

Ratio	Alpha	Beta
Asset turnover	2.5×	1.43×
Equity multiplier	2.0×	3.9×
ROE	23.4%	21.3%
DSO	58 days	73 days
DIO (on COGS)	56 days	130 days
DPO	37 days	47 days
CCC	77 days	156 days
Net debt / EBITDA	1.5×	3.2×
Interest cover (EBIT)	4.7×	2.3×
CFO / EBITDA	81%	33%
FCF	5.5	−2.0

Conclusion: Beta has better margins but ties up far more capital in inventory and receivables, has twice the leverage and burns cash after capex. Alpha's similar ROE comes from efficiency, not leverage. A lender would prefer Alpha; an investor in Beta would need to believe inventory is saleable and that capex will lift returns.

25.4 Earnings quality: warning signs

Signal	Possible issue
Net income growing, CFO flat or falling	Aggressive revenue recognition; working-capital build-up
DSO rising quarter after quarter	Weak customers, channel stuffing, disputed invoices
Inventory growing faster than sales	Obsolescence; over-buying; inflated stock
Capitalised costs rising (development, "other assets")	Expenses shifted to the balance sheet
Frequent "one-off" restructuring charges	Recurring costs presented as exceptional
Large gap between adjusted EBITDA and reported EBIT	Generous adjustments
Changes in auditors, accounting policies or useful lives	Smoothing; disagreement
Related-party sales or purchases	Non-arm's-length transactions
Low effective tax rate vs peers without explanation	Tax risk
Complex structures, frequent acquisitions	Opaque organic growth

PRACTICE · Practice questions

1. Compute DuPont components and ROE: net income $6m, revenue $80m, average assets $60m, average equity $24m.
2. A borrower: EBITDA $20m, interest $4m, scheduled principal $10m, tax paid $2m, maintenance capex $3m. Compute DSCR (cash available = EBITDA − tax − maintenance capex).
3. Revenue +15%, receivables +45%, inventory +30%, net income +20%, CFO −40%. What story would you investigate?

ANSWER · Answers to the practice questions

1. Net margin 7.5%; asset turnover 1.33×; equity multiplier 2.5× → ROE = 0.075 × 1.333 × 2.5 = **25%** (check: 6 ÷ 24 = 25%).

2. Cash available = 20 − 2 − 3 = $15m; debt service = 4 + 10 = $14m → DSCR = **1.07**× — thin; most lenders want ≥ 1.20–1.30×.
3. Earnings are not converting to cash: receivables growing three times faster than sales (extended terms, collection issues, possibly fictitious or premature revenue) and inventory build-up. Investigate customer ageing, top new customers, credit terms granted, returns after period-end, and inventory composition.

CHAPTER 26

Internal Controls and Audit

LEARNING OBJECTIVES · In this chapter

What internal control is, the COSO framework, key controls in the main business cycles, fraud risks, and how external and internal audits work — including what an audit opinion does and does not guarantee.

26.1 Internal control and the COSO framework

Internal control is a process designed to provide reasonable assurance about the achievement of objectives in operations, reporting and compliance. The most widely used framework is COSO's *Internal Control — Integrated Framework* (2013), with five components:

Component	Meaning	Examples
Control environment	Tone at the top, integrity, governance, structure	Code of conduct, independent board, competent finance team
Risk assessment	Identifying and analysing risks to objectives, including fraud	Annual risk assessment, fraud risk review
Control activities	Policies and procedures that mitigate risks	Approvals, reconciliations, segregation of duties, access controls
Information and communication	Quality information flows	Reporting lines, whistleblowing channel
Monitoring activities	Ongoing and separate evaluations	Internal audit, management reviews

Control types: **preventive** (stop errors/fraud: approvals, system validations) vs **detective** (find them: reconciliations, exception reports); **manual** vs **automated** (system-enforced, more reliable if IT general controls are sound).

26.2 Key controls by business cycle

Cycle	Key risks	Key controls
Order-to-cash	Sales to bad credit risks; unbilled deliveries; fictitious sales; misappropriated receipts	Credit limits enforced in ERP; delivery-to-invoice matching; sequential invoice numbers; lockbox/bank collections; receivables reconciliation; credit-note approval
Procure-to-pay	Fictitious suppliers; duplicate payments; overpricing; kickbacks	Vendor master changes controlled and verified by call-back; three-way match; DoA approvals; dual signatures for payments; supplier statement reconciliation
Payroll	Ghost employees; unauthorised changes	HR/payroll segregation; payroll change approvals; headcount reconciliations
Inventory	Theft; obsolescence; misstatement	Physical access control; cycle counts; full year-end count; ageing review
Treasury	Unauthorised payments; fraud; FX losses	Bank mandates; dual authorisation; payment limits; daily bank reconciliation; segregation of dealing and settlement
Financial close	Misstatement	Account reconciliations; review of journals (especially manual, round-sum, period-end); management review
IT general controls	Unauthorised changes or access	User access reviews; change management; backups; segregation of developers and production

COMMON MISTAKES · Classic fraud schemes

- **Business email compromise:** fake email from a "supplier" or "CEO" changes bank details or requests an urgent transfer.
- **Fictitious vendor:** an employee creates a supplier and approves its invoices.
- **Lapping:** stealing one customer's payment and covering it with the next customer's payment.
- **Kickbacks:** purchasing manager receives payments for awarding contracts at inflated prices.
- **Financial statement fraud:** fictitious revenue, hidden liabilities, inventory inflation — often under pressure to meet targets or covenants.
- **Trade fraud:** duplicate financing of the same goods with different banks; fake warehouse receipts.

Fraud triangle (Cressey): **pressure**, **opportunity** and **rationalisation**. Controls reduce opportunity; culture and incentives address the others.

26.3 External audit

An external audit provides **reasonable (not absolute) assurance** that financial statements are free from material misstatement. Audits of most entities follow International Standards on Auditing (ISAs) issued by the IAASB; US public companies are audited under PCAOB standards.

The audit process:

1. **Acceptance and planning:** understand the entity, assess risks, set materiality (often a percentage of profit before tax, revenue or total assets).
2. **Risk assessment:** identify significant risks (revenue recognition and management override of controls are presumed risks under ISA 240).
3. **Tests of controls** where the auditor relies on them.
4. **Substantive procedures:** analytical review; tests of details — confirmations (bank, receivables), attendance at inventory counts, vouching to invoices, recalculation, cut-off tests, review of subsequent events.
5. **Completion:** going-concern assessment, management representations, evaluation of misstatements.
6. **Reporting:** opinion; key audit matters (for listed entities); management letter on control weaknesses.

Opinion	Meaning
Unmodified (clean)	Statements are fairly presented in all material respects
Unmodified with "material uncertainty related to going concern" section	Fair presentation, but significant doubt about continuing as a going concern
Emphasis of matter	Fair, but the auditor draws attention to a disclosed matter
Qualified	Fair except for a specific material (not pervasive) issue
Adverse	Statements are materially and pervasively misstated
Disclaimer	Auditor could not obtain sufficient evidence to form an opinion

PROFESSIONAL INSIGHT · What an audit does not guarantee

A clean audit opinion does not certify that the business is healthy, that management is honest, that there is no fraud below materiality, or that the company will survive. It says the statements are fairly presented under the framework, based on evidence gathered on a sampling and risk basis. Lenders still perform their own analysis; acquirers still do due diligence.

26.4 Internal audit

Internal audit is an independent, objective assurance and consulting function within the organisation, reporting functionally to the board's audit committee. Under the Institute of Internal Auditors' framework (Global Internal Audit Standards, effective January 2025), it evaluates governance, risk management and control processes. In the **three lines model**: management owns and manages risk (first line), risk and compliance functions provide expertise and oversight (second line), and internal audit provides independent assurance (third line).

PRACTICE · Practice questions

1. A purchasing manager can create suppliers in the ERP, approve purchase orders up to $100k and approve invoices for payment. Identify the risk and propose controls.
2. Name three audit procedures to verify trade receivables.
3. The auditor's report includes a "material uncertainty related to going concern" section. You are a supplier asked to give 90-day credit. What do you do?

ANSWER · Answers to the practice questions

1. Lack of segregation of duties: he could create a fictitious supplier, raise POs and approve payment. Controls: vendor master creation by a separate team with independent verification (registration, bank letter, call-back); invoice approval by budget owner and AP; three-way match; periodic review of new suppliers and payments; rotation and audit analytics (duplicate bank accounts between employees and vendors).
2. External confirmation of balances with customers; testing subsequent receipts after year-end; reviewing the ageing and allowance; cut-off testing of invoices and delivery notes around year-end; vouching sample balances to contracts and proofs of delivery.
3. Treat as high risk: request recent management accounts and cash forecast, reduce terms (advance payment or cash against documents), require a bank guarantee/SBLC or credit insurance cover, set a low credit limit, and monitor payments closely.

SUMMARY · Part III summary

- Double entry and the accounting equation make the three statements mutually consistent; master the three-statement effects of every common transaction.
- Revenue follows transfer of control (IFRS 15); expenses follow consumption of benefits; this creates the timing gap with cash.
- Balance-sheet judgements — inventory, receivables, useful lives, provisions, leases — drive the quality of reported numbers.
- Management accounting turns data into decisions: contribution, relevant costs, budgets and variances.
- Analysis is about drivers and cash conversion, not computing every ratio.
- Controls and audits reduce, but never eliminate, the risk of error and fraud.

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